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Eigenmodes of brain activity: Neural field theory predictions and comparison with experiment

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ABSTRACT

Neural field theory of the corticothalamic system is applied to predict and analyze the activity eigenmodes of the bihemispheric brain, focusing particularly on their spatial structure. The eigenmodes of a single brain hemisphere are found to be close analogs of spherical harmonics, which are the natural modes of the sphere. Instead of multiple eigenvalues being equal, as in the spherical case, cortical folding splits them to have distinct values. Inclusion of interhemispheric connections between homologous regions via the corpus callosum leads to further splitting that depends on symmetry or antisymmetry of activity between brain hemispheres, and the strength and sign of the interhemispheric connections. Symmetry properties of the lowest observed eigenmodes strongly constrain the interhemispheric connectivity strengths and unihemispheric mode spectra, and it is predicted that most spontaneous brain activity will be symmetric between hemispheres, consistent with observations. Comparison with the eigenmodes of an experimental anatomical connectivity matrix confirms these results, permits the relative strengths of intrahemispheric and interhemispheric connectivities to be approximately inferred from their eigenvalues, and lays the foundation for further experimental tests. The results are consistent with brain activity being in corticothalamic eigenmodes, rather than discrete "networks" and open the way to new approaches to brain analysis.

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Introduction

It has been widely observed that "resting state" brain activity has distinct patterns, often termed "resting state networks" (RSNs) (Smith et al., 2009; Damoiseaux et al., 2006). Typically 5–10 patterns are seen and these prove to be very similar to the activity patterns excited during task performance (Smith et al., 2009; Damoiseaux et al., 2006; Mantini et al., 2007; De Luca et al., 2006). Moreover, these patterns can depend on frequency and exhibit low-frequency connectivity fluctuations themselves (Robinson, 2003; Kitzbichler et al., 2009; Shen et al., 1999; Hipp et al., 2012; Cabral et al., 2011), likely reflecting critical fluctuations and frequency dependences of activity correlation lengths (Robinson, 2003, 2012). Similarly, the "default mode network" (DMN) is a set of brain regions that increase their activity in the resting state relative to task states, especially goal-directed ones (Hagmann et al., 2008; Raichle et al., 2001; Gusnard and Raichle, 2001; Fransson, 2005, 2006).

The term "resting state" is a misnomer because the normal brain is always active and processing incoming signals, even if not consciously. Indeed, RSNs are similar in wake, sleep, and even anesthesia (Smith et al., 2009; Vincent et al., 2007; Ghosh et al., 2008; Gusnard and Raichle, 2001), with constant processing of incoming signals from the nervous system, maintenance of homeostasis, memory consolidation, and other processing, whether or not these are induced by an experimenter ("tasks"). Moreover, there are divergent definitions of what is termed "activity" — some authors base this on metabolic load, as measured by positron emission tomography (PET), for example; others define it to be proportional to power in fluctuations in band-limited hemodynamic signals, as detected by functional magnetic resonance imaging (fMRI) or electroencephalography (EEG) (Raichle et al., 2001; Gusnard and Raichle, 2001; Mehrkanoon et al., 2014). Whatever definition is used, neural activity ultimately drives the relevant signals.

RSNs, the DMN, and other brain "modes" (a term that appears to have no generally agreed, precise definition in the neuroscience literature) are often pictured as being substantially segregated networks that either exist in the long term, or are formed temporarily to process particular inputs. Indeed, much effort has been expended on trying to identify specific cognitive functions underpinned by each RSN. From this perspective it is paradoxical that very similar patterns of activity are seen in sleep and anesthesia when the subject is unconscious (Vincent et al., 2007; Ghosh et al., 2008; Fukunaga et al., 2006;

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Hutchison et al., 2013), which has thrown into question the notion of RSNs being purely related to tasks or cognition. Notably, RSNs are most often derived via independent component analysis (ICA) and closely related methods. These mathematical signal analysis methods do not account for the nature of the system that produces the signals, and are thus not based on brain physiology. Instead they extract characteristic orthogonal components of the signal that reflect independent contributions to the non-Gaussian part of its statistics; these components thus simultaneously incorporate aspects of both the input patterns and the brain eigenmodes. Often components are further processed to produce intrinsically positive statistical quantities, such as z-scores that do not record whether specific "networks" have higher or lower activity than baseline. An approach that retains physiological and activity-related information has potential to provide more insight than these types of blind signal analysis.

A network-based picture of brain activity can be founded on the fact that neurons do form a discrete network at microscopic scales. However, the "networks" discussed in the previous paragraphs are based on brain activity and connectivity on much coarser scales — typically with fewer than 1000 nodes, or regions of interest (RoIs), covering the whole brain with its 10^{11} neurons, and thus typically over 10^8 neurons per node. This poses serious problems for any picture of brain activity being in networks that are largely segregated from one another:

(i) It is well established that the typical number of steps separating any pair of RoIs is only around 3 when the total number of RoIs is of order 1000 (Bullmore and Sporns, 2009; Henderson and Robinson, 2013, 2014), so subnetworks of these coarse-grained networks are actually closely integrated. This is also true at the microscopic neural level, where each cortical neuron receives inputs from around 10^4 others and projects to a similar number, with white-matter projections having a mean range of several cm — far beyond the bounds of a typical RoI or local circuits (Nunez, 1995; Henderson and Robinson, 2013, 2014; Robinson et al., 2004, 1997). Moreover, since the system is near to a critical point, the correlation length encompasses the whole brain and activity extends across scales, including via indirect corrections that link all areas more closely than the direct connections alone (Robinson et al., 1997, 2014; Beggs and Plenz, 2003; Bassett et al., 2006; Robinson, 2003, 2012). Note that the system is not at the critical point, so linear analysis can still be used.

(ii) Point (i) implies that there is a prima facie problem with viewing a network as having discrete connections between a relatively limited number of nodes — these will only be average connectivities, and many if not most influences they approximate actually travel indirectly through a multitude of polysynaptic paths that are not shown directly in matrices of connections between macroscopic RoIs (Robinson, 2012, 2013; Robinson et al., 2014; Galán, 2008). Indeed, it has been estimated that around 90% of activity in the cortex is internally generated, rather than being directly stimulated by an external input (Robinson et al., 2014, 1997).

(iii) With around 10^8 neurons and 10^{12} synapses per RoI, the underlying neural tissue is approximately a continuum on the scale of the RoIs.

(iv) Unless the number of RoIs is sufficient to resolve all the scales relevant to a given problem, connectivity patterns will depend on the resolution of the representation, with coarser-grained connection matrices having a higher fraction of nonzero entries (Cammoun et al., 2012; Zalesky et al., 2010). Networks of connections between RoIs, especially fMRI-derived functional networks, are thus not the actual brain networks. Rather they are networks of effective links or correlations between points on a relatively sparse set of observation points.

We recently showed that sparse networks of RoIs can yield useful information on the dynamics of brain activity by sampling the lowest order eigenmodes of the underlying brain network, provided these eigenmodes dominate the dynamics (Robinson, 2013). Connection matrices (CMs) then measure the effective coupling between observation points via activity in these low-order eigenmodes. That work highlighted the need to make a careful choice of sampling points if one is to reduce the dynamics to effective dynamics on a sparse network of only a few RoIs, as is usually done a priori in dynamic causal modeling (DCM) (for example Friston et al., 2003; Seghier and Friston, 2013; Robinson, 2013; Friston, 2011). Currently, most work on understanding brain connectivity and activity patterns proceeds by means of statistical or dynamical models of networks of effective interactions between time series recorded at nodes of a sensor network and/or via anatomical tracing of links between points on specific networks of various resolutions. Here we pursue the complementary and more fundamental approach of developing a theory of how underlying brain activity gives rise to wide classes of measurements on whatever sensor network is used in a particular situation. A comprehensive theory of this type would allow all the observables and their interdependences to be calculated from the underlying dynamics, and strengthen the ability to make inferences in the reverse direction.

Neural field theory (NFT) has been particularly successful in analyzing medium-to-large scale brain activity — from about 0.1 mm to the whole brain. This approach averages neural quantities such as firing rates, soma voltages, and incoming activity levels over multiple neurons to obtain equations for their evolution on scales of a few tenths of a mm up (Deco et al., 2008). The resulting NFTs have been successfully used to explain a wide variety of brain activity phenomena and structure–function relationships (Deco et al., 2008; Jirsa and Haken, 1996; Nunez, 1995; Robinson et al., 1997; Robinson, 2003; Buice and Cowan, 2007; Robinson, 2005; Robinson et al., 2002; Breakspear et al., 2006; Freeman, 1975; Nunez, 1974; Beurle, 1956; Wilson and Cowan, 1973; Steyn-Ross et al., 2005; Wright and Liley, 1996; Robinson et al., 2005; Robinson et al., 2001b; Ermentrout, 1998; Coombes, 2005; Amari, 1977; Bressloff, 2002; Bressloff and Cowan, 2002; Ermentrout and Cowan, 1979; Lopes da Silva et al., 1974; Robinson, 2012).

One area of application of NFT has been to the corticothalamic system, which is chiefly responsible for the generation of EEG and fMRI signals, and many other observable linear and nonlinear brain phenomena (Robinson et al., 2002, 2005; Robinson, 2005). However, one aspect that has not yet been incorporated explicitly is the bihemispheric structure of the brain, in which homologous regions (located symmetrically relative to the sagittal midplane of the brain) are connected approximately one-to-one via axons running through the corpus callosum and subsidiary structures (Kandel et al., 2000; Quigley et al., 2003; Tyszka et al., 2011). It is notable that most brain activity is observed to be symmetric between hemispheres, but it is not known whether this is due to inputs being predominantly symmetric, to symmetric responses being the most easily excited, or to both effects.

The strongest EEG signals are in the lowest spatial eigenmodes, with a large fraction of EEG power being in the global eigenmode, which is symmetric between hemispheres and approximately uniform across each (Nunez, 1995; Nunez and Srinivasan, 2006; Robinson et al., 2001a). Theory indicates that the globally uniform eigenmode is the least stable (i.e., the least damped), and hence the most easily excited (Robinson, 2003; Robinson et al., 1997, 2001a, 2014), at least for the class of NFTs considered here, which represent connectivities that fall off smoothly with distance. More generally, once activity has been calculated in NFT it can be used to compute other quantities such as the correlations used to define functional connection matrices (fCMs), and to demonstrate the identity of the eigenmodes of activity, anatomical CMs, and fCMs (Robinson, 2012; Robinson et al., 2014). Since all activity can be expressed in terms of the physical eigenmodes of the system, these provide a systematic basis for wider-ranging analyses, computation of quantities such as RSNs, and conceptual and practical advances more generally. Notably, eigenmodes differ in a critical way from the components inferred via methods like ICA and PCA: eigenmodes are entities that derive from the dynamics of the brain as the physical system that generates observed activity, and their nature directly constrains brain anatomy and physiology; in contrast, components are statistically independent entities that are obtained from time series without regard

to the nature of the system that generates them or the physical quantities they represent. Both types of entities are useful, but this fundamental difference must always be borne in mind. We stress that eigenmodes computed from theory or observation are approximations to the true physical eigenmodes of the system.

This paper's key aims are thus to: (i) find the NFT eigenmodes of a brain hemisphere and of the bihemispheric brain with transcallosal coupling (Robinson et al., 2015), particularly focusing on their spatial structure; (ii) analyze the eigenmodes and eigenvalues of a single brain hemisphere, and examine how coupling between hemispheres changes the eigenvalue spectrum; and (iii) show that these eigenmodes are consistent with those of unihemispheric and bihemispheric brain connectivity matrices obtained from MRI.

Methods

In this section we outline the methods used to describe curved cortical surfaces and visualize them on spherical and planar projections. We describe neural field theory (NFT) and how to solve NFT equations on these surfaces to obtain eigenmodes, and briefly summarize the established NFT that we apply to describe the corticothalamic system. Published experimental anatomical connection matrices (aCMs) that provide test data are also briefly discussed.

Inflation of cortical surface into a sphere for visualization and to investigate intermediate shapes

The shape of the cortical surface, and the patterns of activity it supports, are generally hard to visualize because folding hides regions that are located in sulci, and the "far" side of the brain is completely hidden from any single viewpoint.

To expose sulcal regions, many authors have "inflated" a brain hemisphere onto a sphere, since they are topologically equivalent aside from where the corpus callosum and other smaller interhemispheric connections pass through, via ~0.5% of the total area (Kandel et al., 2000). The most widely used methods to do this minimize distortions of the surface during inflation by introducing a cost function that rewards uniformity of curvature, but penalizes distortions of the geometry (Fischl et al., 1999; Seo and Chung, 2011; Seo et al., 2010; Lehoucq et al., 1998; Chung, 2002); this function is minimized at each step of inflation. Ultimately, this process maps each point r of a cortical hemisphere onto a sphere at a point (θ, φ), where θ and φ are the standard spherical coordinates of polar angle (measured from the north pole) and azimuthal angle, respectively. In the present work, we use the inflation algorithm implemented in the Freesurfer software (Fischl et al., 1999).

The above method of unfolding the cortex onto a sphere establishes a one-to-one mapping between the two and naturally generates a family of partly unfolded surfaces that are intermediate in shape between the cortex and the corresponding sphere. When we solve our neural field equations for brain activity on these surfaces, this enables us to trace the eigenmodes and eigenvalues continuously as the dynamics change from the fully folded case to the spherical case, where the eigenmodes are spherical harmonics whose properties are well understood (Olver et al., 2010). This proves to be particularly useful for understanding the correspondence between cortical eigenmodes and spherical harmonics. Additionally, we hold the cortical area constant as it is inflated, which is done post-hoc by calculating the area at each stage of inflation and rescaling the linear dimensions by the factor needed to keep the area constant.

Flattening of cortical surface onto a plane for visualization

Even on a sphere, eigenmodes of brain activity can be hard to visualize, since when the sphere is embedded in three-dimensional space there is always a far side, whichever direction it is viewed from. Hence, we flatten the sphere onto a plane for visualization. We use the simple sinusoidal equal-area projection

$$x = \varphi \sin\theta, \tag{1}$$

$$y = \frac{\pi}{2} - \theta, \tag{2}$$

which preserves the 4π area of the unit sphere. The north pole ($\theta = 0$) and south pole ($\theta = \pi$) are mapped to $(x,y) = (0,\pi/2)$ and $(0,-\pi/2)$, respectively, while the equator $(\theta,\pi/2)$ has $y = 0$ and $-\pi < x \leq \pi$.

In the left hemisphere of the brain, we use the coordinate system just described, with the z axis pointing vertically upward, the x axis toward the right hemisphere in the horizontal plane, and the y axis pointing toward the front of the head. In the right hemisphere, the situation is the same except that mirror symmetry forces us to place the x axis pointing toward the left hemisphere. This means that homologous structures are assigned the same θ and φ in a perfectly symmetric brain.

Neural field theory of brain eigenmodes

Neural field theory (NFT) has been widely discussed elsewhere (Deco et al., 2008; Jirsa and Haken, 1996; Nunez, 1995, 1974; Robinson et al., 1997, 2002, 2005, 2001b; Robinson, 2003, 2005; Buice and Cowan, 2007; Breakspear et al., 2006; Freeman, 1975; Beurle, 1956; Wilson and Cowan, 1973; Steyn-Ross et al., 2005; Wright and Liley, 1996; Ermentrout, 1998; Coombes, 2005; Amari, 1977; Bressloff, 2002; Bressloff and Cowan, 2002; Ermentrout and Cowan, 1979; Lopes da Silva et al., 1974), including its successful application to numerous brain activity phenomena. NFT averages neural quantities such as firing rates, cell-body voltages, and levels of incoming/outgoing activity carried by axons over multiple neurons to obtain equations for the evolution of these averages (and higher moments in some analyses) on scales of a few tenths of a mm and above.

In the present work, we employ NFT both at a general level described in this subsection and in the particular corticothalamic system that is primarily responsible for driving the signals observed in EEG and fMRI (see the Corticothalamic NFT of one hemisphere section).

In a wide class of NFTs, the dynamics of the average brain activity $\phi^H(\boldsymbol{r},t)$ on a brain hemisphere $H = L,R$, with R denoting right and L left, can be written in terms of a partial differential equation that is derived from average dendritic, somatic, and axonal physiology. NFT equations of this form can always be written in integral form by use of a Green-function representation (Robinson, 2012); the converse does not apply, but we do not consider other types of NFT here.

Here we assume that the underlying brain connectivity is time-independent on the scales of the activity variations, so the connectivity between points depends only on their positions. In general, we note that connectivity includes time delays for signals to travel between regions (Robinson, 2012). The use of a time-independent connectivity does not prevent inclusion of activity-dependent effects such as facilitation and habituation; these are treated as part of the response in this approach, but can equally well be interpreted as dynamic connectivity changes; e.g., due to changes in synaptic strength or spiking threshold (Robinson and Roy, 2015). Both approaches are equivalent, but the present approach avoids evaluation of the other responses on a changing network — instead, all responses, including network connectivity changes, are computed relative to an underlying mean network structure that is fixed on the relevant timescales. Development, learning, and similar changes take place on much longer timescales and are not considered here.

A further approximation we make is to examine only moderate changes relative to baseline, so that responses can be treated in the linear approximation. As noted in the Introduction, this has been very successful in predicting phenomena such as EEG spectra, evoked potentials, and the blood oxygen level dependent (BOLD) response in fMRI (Robinson, 2012; Rennie et al., 2002; Kerr et al., 2011; Drysdale et al.,

2010; Deco et al., 2008; Aquino et al., 2014), but does not cover dynamics of epileptic seizures, intense steady state visual evoked potentials, bistability of intense alpha rhythms, or similar nonlinear phenomena (Robinson et al., 2002; Breakspear et al., 2006; Roberts and Robinson, 2012; Freyer et al., 2009, 2011). In the present context, we note that the fMRI and electroencephalographic (EEG) signals that are usually used to infer connectivities are usually near-linear perturbations on normal steady states, even where those steady states are determined by nonlinear interactions within the brain, a point we discussed at more length previously (Robinson, 2012; Robinson et al., 2014). These signals primarily reflect activity afferent on pyramidal neurons (Nunez, 1995; Robinson et al., 2001a; Niedermeyer and Lopes da Silva, 1999). Moreover, in the cortex, the only long-range connections are via pyramidal white-matter axons, with other interactions primarily modulating activity through short-range influences.

Given the above points, we can temporally Fourier transform the dynamical equations to convert temporal derivatives into algebraic form, with an implicit time dependence $e^{-i\omega t}$, where ω is the angular frequency. Then NFT yields a partial differential equation of the form

$$\mathcal{D}(\boldsymbol{r},\omega)\phi^H(\boldsymbol{r},\omega) = C(\boldsymbol{r},\omega)\phi_x^H(\boldsymbol{r},\omega), \tag{3}$$

where $\boldsymbol{r}$ is position on the cortex, $\mathcal{D}$ is a spatial differential operator, ϕ_x^H is a field of external inputs that drive the internal activity ϕ^H, and C is a coupling coefficient (Robinson, 2014). The form (3) can accommodate the involvement of subcortical structures, as we see in the next subsection. This is important because the thalamus, for example, has been implicated in the dynamics of RSNs (Mantini et al., 2007) and brain activity more generally (Robinson et al., 2002, 2005; Breakspear et al., 2006; Robinson, 2005; Deco et al., 2008).

Cortical hemispheres are coupled approximately one-to-one between homologous regions, as seen in Fig. 1. If we label these regions with the same $\boldsymbol{r}$ in each case, and assume uniform one-to-one coupling $I(\omega)$ as a first approximation, we can write

$$\mathcal{D}(\boldsymbol{r},\omega)\phi^R(\boldsymbol{r},\omega) = I(\omega)\phi^L(\boldsymbol{r},\omega) + C(\boldsymbol{r},\omega)\phi_x^R(\boldsymbol{r},\omega), \tag{4}$$

$$\mathcal{D}(\boldsymbol{r},\omega)\phi^L(\boldsymbol{r},\omega) = I(\omega)\phi^R(\boldsymbol{r},\omega) + C(\boldsymbol{r},\omega)\phi_x^L(\boldsymbol{r},\omega). \tag{5}$$

The terms $I(\omega)$ describe the strengths, delays, and frequency dependences of the interhemispheric couplings. For simplicity, the two hemispheres and interhemispheric couplings have been assumed identical in Eqs. (4) and (5).

Corticothalamic NFT of one hemisphere

Cortical responses are modulated by corticothalamic connections, which affect the propagation of activity across the cortex, especially by inducing resonances such as the alpha and beta rhythms (Deco et al., 2008; Robinson et al., 2002). We thus apply an established model of the corticothalamic system to a single hemisphere, schematically illustrated in either half of Fig. 2 (Deco et al., 2008; Robinson et al., 2004, 2005, 2001b, 2002; Robinson, 2005; Breakspear et al., 2006). Other variants are certainly possible, but the one chosen suffices for our purposes.

The corticothalamic model includes cortical excitatory (e) and inhibitory (i) populations, along with populations representing thalamic specific relay (s) nuclei and the thalamic reticular (r) nucleus; external inputs (n) are also included (Robinson et al., 2002; Deco et al., 2008; Robinson, 2005). It has sufficient flexibility to model corticothalamic systems using physiologically realistic parameters, while also enabling specialized limiting cases to be examined to elucidate fundamental issues (Deco et al., 2008; Jirsa and Haken, 1996; Nunez, 1995, 1974; Robinson et al., 1997, 2002; Robinson, 2003, 2005; Buice and Cowan, 2007; Breakspear et al., 2006; Freeman, 1975; Beurle, 1956; Wilson and Cowan, 1973; Steyn-Ross et al., 2005; Wright and Liley, 1996). The model discussed here has been extensively tested and verified in its linear and nonlinear forms against a wide range of experiments on corticothalamic phenomena and its physiological parameters have been constrained by experiment (Deco et al., 2008; Robinson et al., 2004, 2005; Robinson, 2005).

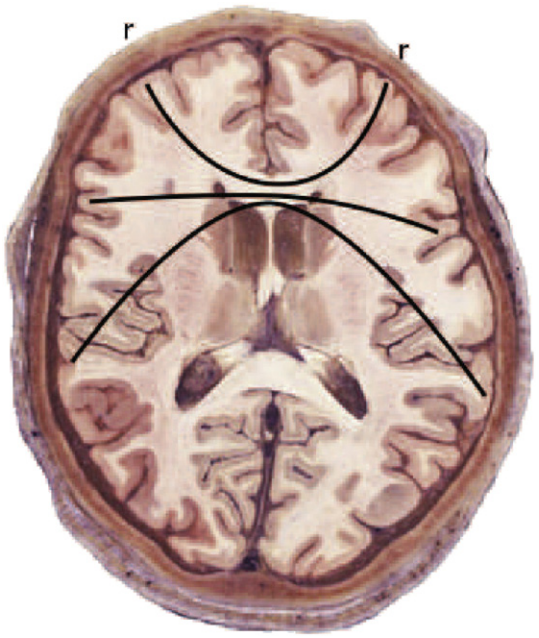


Fig. 1. Schematic of interhemispheric connections and points mapped to the same position coordinate $\boldsymbol{r}$. Because of the near-symmetry of the brain under reflection across the midplane of the head, points with the same $\boldsymbol{r}$ in opposite hemispheres have mirror-image locations. Adapted from Mai et al. (2008).

In our model, we approximate the cortex as a two-dimensional sheet, owing to its thinness, while the coordinates in the thalamus are related one-to-one to those in the cortex via the primary thalamocortical mapping, with a single coordinate used to refer to corresponding points in all populations and structures (Robinson, 2005). The specific NFT and notation are those of Robinson (2005), so we simply summarize the main features and key results here.

It has been established that the system corresponding to one hemisphere in Fig. 2 has a steady state solution with low population firing rates, so long as the input is not too strong (Robinson et al., 2004, 1997, 2002, 2005, 2001b). As discussed above, we consider small perturbations from this fixed point.

Perturbations Q_a ($a = e,i,r,s$) of mean firing rate can be approximated as being linearly related to perturbations of the mean soma voltage V_a, with

$$Q_a(\boldsymbol{r},t) = \rho_a V_a(\boldsymbol{r},t), \tag{6}$$

where ρ_a is the slope of the full nonlinear response curve that relates Q_a to V_a, evaluated at the fixed point; ρ_a is assumed constant here.

The perturbation V_a is a sum of contributions V_{ab} from various populations b.

$$V_a(\boldsymbol{r},t) = \sum_b V_{ab}(\boldsymbol{r},t) \tag{7}$$

The subpotential V_{ab} responds to the incoming perturbation field of afferent activity ϕ_{ab} to neurons a from neurons b according to

$$\mathcal{D}_{ab}(t)V_{ab}(\boldsymbol{r},t) = \nu_{ab}\phi_{ab}(\boldsymbol{r},t-\tau_{ab}), \tag{8}$$

$$\mathcal{D}_{ab}(t) = \frac{1}{\alpha_{ab}\beta_{ab}}\frac{d^2}{dt^2} + \left(\frac{1}{\alpha_{ab}} + \frac{1}{\beta_{ab}}\right)\frac{d}{dt} + 1, \tag{9}$$

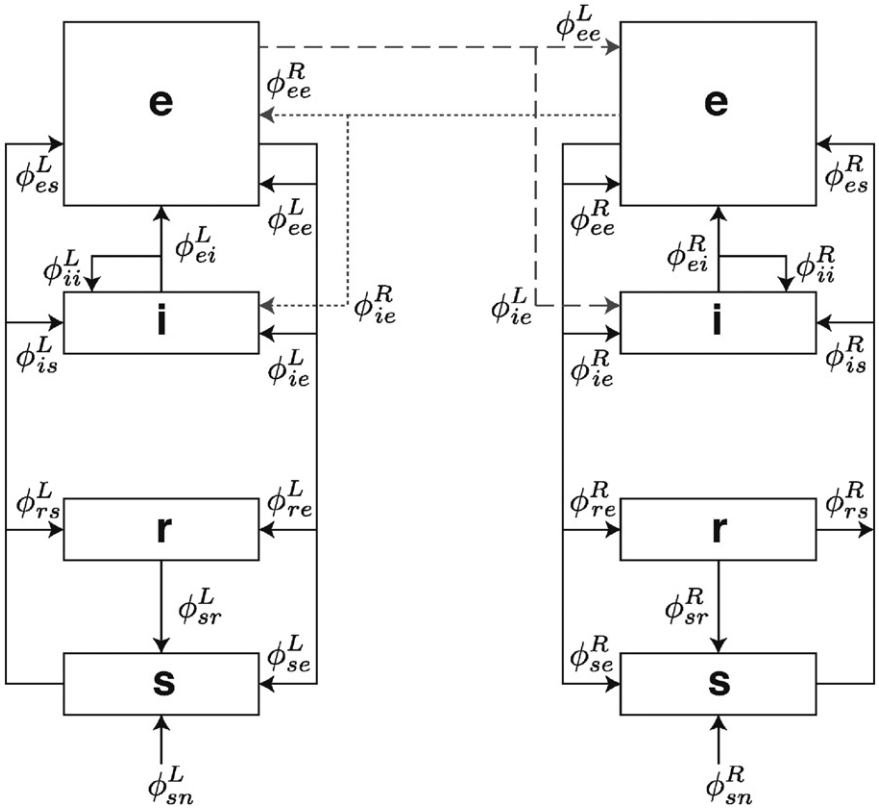


Fig. 2. Schematic of the bihemispheric corticothalamic system considered, showing the excitatory (e), inhibitory (i), reticular nucleus (r), and specific nuclei (s) populations, inputs n and the fields ϕ_{ab}^{H} that travel between them, as discussed in detail in the text. Superscripts L and R indicate left and right brain hemispheres, respectively.

where α_{ab} and β_{ab} are characteristic rates of soma voltage decay and rise, respectively, which we treat as independent of a and b here, although this assumption can be easily relaxed (Robinson, 2005). The afferent populations include external inputs n. In Eq. (8)

$$\nu_{ab} = N_{ab} s_{ab}, \tag{10}$$

N_{ab} is the average number of synapses on each neuron of type a from neurons of type b, s_{ab} is the synaptic strength expressed as the time-integrated soma voltage response to a unit input, and τ_{ab} is any discrete time delay required for signals to reach population a from b and is always zero if $a = b$ (for $a = b$ time delays are accounted for via the internal propagation of ϕ_{aa}, as discussed below). Quantities such as velocities, ranges, and time delays are interpreted as weighted average, or effective, values in NFT.

The mean activity fields ϕ_{ab} derive from the source Q_b, and are approximated as propagating according to a damped wave equation. A number such equations have been proposed (Deco et al., 2008; Robinson et al., 1997; Jirsa and Haken, 1996; Beurle, 1956; Nunez, 1995, 1974; Steyn-Ross et al., 2005; Wright and Liley, 1996). One such form is (Robinson et al., 1997; Robinson, 2005)

$$\mathcal{D}_{ab}(\mathbf{r}, t)\phi_{ab}(\mathbf{r}, t) = Q_b(\mathbf{r}, t), \tag{11}$$

$$\mathcal{D}_{ab}(\mathbf{r}, t) = \frac{1}{\gamma_{ab}^2}\frac{\partial^2}{\partial t^2} + \frac{2}{\gamma_{ab}}\frac{\partial}{\partial t} + 1 - r_{ab}^2\nabla^2, \tag{12}$$

where r_{ab} is the characteristic range of axons projecting to neurons a from b, v_{ab} is the mean axonal velocity, and $\gamma_{ab} = v_{ab}/r_{ab}$ is an effective damping rate that reflects the rate at which spikes traverse the finite range of axons to reach axonal terminals. The approximate isotropy and roughly exponential axonal range distribution implicit in Eq. (12) have been supported by multiple studies, ranging from anatomical investigations (Braitenberg and Schüz, 1991) to imaging (Henderson and Robinson, 2013). Other equations can be used in place of Eqs. (11) and (12) to incorporate asymmetries, axonal velocity dispersion, and other effects (Robinson et al., 1997; Nunez, 1995; Jirsa and Haken, 1996; Bojak and Liley, 2010; Roberts and Robinson, 2008), with corresponding modifications to what follows, but the above form suffices for our present analysis and has been successfully applied to a wide range of corticothalamic phenomena, as outlined above.

If we Fourier transform the above equations vs. time, we find

$$Q_a(\mathbf{r}, \omega) = \rho_a V_a(\mathbf{r}, \omega), \tag{13}$$

$$V_a(\mathbf{r}, \omega) = \sum_b V_{ab}(\mathbf{r}, \omega), \tag{14}$$

$$V_{ab}(\mathbf{r}, \omega) = L(\omega)\nu_{ab}\phi_{ab}(\mathbf{r}, \omega)e^{i\omega\tau_{ab}}, \tag{15}$$

$$L_{ab}(\omega) = (1 - i\omega/\alpha_{ab})^{-1}(1 - i\omega/\beta_{ab})^{-1}, \tag{16}$$

$$\mathcal{D}_{ab}(\mathbf{r}, \omega)\phi_{ab}(\mathbf{r}, \omega) = Q_b(\mathbf{r}, \omega) \tag{17}$$

where Eq. (16) defines the response function $L_{ab}(\omega)$. For convenience in the discussion below, we also define the quantities

$$J_{ab}(\omega) = G_{ab}L_{ab}(\omega)e^{i\omega\tau_{ab}}, \tag{18}$$

$$G_{ab} = \rho_a \nu_{ab}, \tag{19}$$

where G_{ab} is the gain associated with responses in population a due to signals in population b.

In the present analysis we assume parameter values and connectivities in accord with our previous work (Robinson, 2005), where we discussed their justification in detail. It has been found that $r_{ab} \approx 0$ is a reasonable approximation whenever $b = i, r, s$, implying that the

corresponding γ_{ab} can be approximated as infinite in Eq. (12), which gives the local interaction approximation in which $D_{ab}(\boldsymbol{k},\omega)=1$ for these populations (Robinson et al., 1997; Robinson, 2005); connections between different $\boldsymbol{r}$ are then due to white matter fibers from excitatory cortical pyramidal neurons. The only nonzero τ_{ab} are $\tau_{es}=\tau_{se}=\tau_{re}=t_0/2$, where $t_0\approx 85$ ms is the mean axonal time delay involved in a signal traveling from cortex to thalamus and back; intrapopulation and short-range axonal delays are included via the γ_{ab} (Robinson et al., 2002, 2004; Robinson, 2005). Since the average number of synapses linking cortical neurons of types a and b is closely proportional to the numbers of neurons involved (Wright and Liley, 1996; Braitenberg and Schüz, 1991; Robinson et al., 1997), and synaptic properties (e.g., excitatory or inhibitory) primarily depend on the afferent neuron, we can make the random connectivity approximation that $G_{ib}=G_{eb}$ for $b=e,i,s$ (Wright and Liley, 1996) and we further assume $L_{ib}=L_{eb}$ and $\tau_{ib}=\tau_{eb}$, so $J_{ib}=J_{eb}$. Here and in much of what follows, we often omit the arguments $\boldsymbol{r}$ and ω for compactness.

If we use the above approximations and eliminate the variables other than ϕ_{ee} and the input ϕ_{sn} from the above equations, we find (Robinson, 2005, 2012)

$$\left[\mathcal{D}_{ee}-\frac{J_{ee}}{1-J_{ei}-J_{ctc}}\right]\phi_{ee} = \frac{J_{es}J_{sn}}{(1-J_{sr}J_{rs})(1-J_{ei}-J_{ctc})}\phi_{sn}, \tag{20}$$

$$J_{ctc}=\frac{J_{es}(J_{se}+J_{sr}J_{re})}{1-J_{sr}J_{rs}}, \tag{21}$$

$$\mathcal{D}_{ee}=(1-i\omega/\gamma_{ee})^2-r_{ee}^2\nabla^2, \tag{22}$$

where J_{ctc} is the overall corticothalamic loop coupling and the arguments $\boldsymbol{r}$ and ω have been omitted for compactness. For simplicity, we will also use the notation

$$(\mathcal{D}-\mathcal{J})\phi_{ee}=C\phi_{sn}, \tag{23}$$

when no confusion arises, with $\mathcal{D}$, $\mathcal{J}$, and C defined by one-to-one correspondence between terms in Eqs. (20) and (23). The above points show that our corticothalamic NFT can be written in the form (3) in the linear case, with the correspondences

$$\phi^H\leftrightarrow\phi_{ee},\ \phi_x^H\leftrightarrow\phi_{sn}, \tag{24}$$

$$D(r,\omega)\leftrightarrow\mathcal{D}-\mathcal{J},\ C(r,\omega)\leftrightarrow\mathcal{C}. \tag{25}$$

Solution of the Helmholtz equation on cortical hemisphere to obtain NFT eigenmodes

In this section we show how to solve the above NFT equations to obtain eigenmodes on a convoluted brain hemisphere in the case in which the only nonuniformities in brain properties are due to its geometry. If this approximation is not satisfied, numerical solutions are likely to be most convenient; however, approximate first-order uniformity of cortical structure is well established (Kandel et al., 2000) and brain connectivity is also surprisingly similar at various points (Henderson and Robinson, 2013, 2014; Braitenberg and Schüz, 1991).

In the absence of external inputs, (20) can be rearranged into the form

$$D'(\omega)\phi_{ee}=r_{ee}^2\nabla^2\phi_{ee}, \tag{26}$$

$$D'(\omega)=\left(1-\frac{i\omega}{\gamma_{ee}}\right)^2-\frac{J_{ee}}{1-J_{ei}-J_{ctc}}. \tag{27}$$

We next make the approximation that the J_{ab} are spatially uniform, although they still depend on ω, so D' does not have any $\boldsymbol{r}$-dependence. This approximation can easily be relaxed if the eigenfunctions are calculated numerically, which will usually be the situation in practical applications, but we do not do so in the analytic work here because our focus is on geometric nonuniformities rather than parametric ones. At a given frequency, we can now introduce the ansatz

$$\phi_{ee}=y_\eta(\boldsymbol{r})e^{-i\omega_\eta t}, \tag{28}$$

for a spatial eigenmode $y_\eta(\boldsymbol{r})$ with eigenfrequency ω_η, where η labels the eigenmode. Substituting this into Eq. (26) and rearranging the result gives

$$\frac{\nabla^2 y_\eta(\boldsymbol{r})}{y_\eta(\boldsymbol{r})}=\frac{D'(\omega)}{r_{ee}^2}. \tag{29}$$

Since the left of this equation is independent of ω and the right is independent of $\boldsymbol{r}$, we can set both sides equal to a separation constant $-k_\eta^2$. This yields

$$\nabla^2 y_\eta(\boldsymbol{r})=-k_\eta^2 y_\eta(\boldsymbol{r}), \tag{30}$$

$$D'(\omega_\eta)=-k_\eta^2 r_{ee}^2. \tag{31}$$

Eq. (30) must be solved on the curved cortical surface, thereby selecting a discrete family of eigenmodes and corresponding eigenvalues. Once the eigenvalue is known, it is substituted into the dispersion relation (31) to obtain the eigenfrequency, although we focus mostly on the spatial structure in the present work. Dimensional analysis implies that the eigenvalue k_η^2 scales inversely with the area of the surface. Hence, aside from their linear scale, the spatial structure of the eigenmodes does not depend on the linear size of the cortex. Moreover, continuity of the eigenfunctions and their derivatives on the closed cortical surface implies that $k_\eta^2\geq 0$.

The operator ∇^2 above is the Laplace–Beltrami operator, which generalizes the Laplacian to a curved surface (Fischl et al., 1999; Seo and Chung, 2011):

$$\nabla^2 f=\sum_{i,j}\frac{1}{|g|^{1/2}}\frac{\partial}{\partial x^i}\left(|g|^{1/2}g^{ij}\frac{\partial f}{\partial x^j}\right), \tag{32}$$

where the x^j are generalized coordinates that parametrize the surface, g is the metric tensor, and $|g|$ is its determinant.

To obtain the eigenmodes and eigenvalues of Eq. (30) on a curved cortex we use a finite-element discretization based on a recent formulation (Seo and Chung, 2011; Seo et al., 2010; Fischl et al., 1999; Lehoucq et al., 1998; Chung, 2002), which implements the implicitly restarted Arnoldi method on a cortical mesh (Seo and Chung, 2011; Lehoucq et al., 1998; Chung, 2002), as implemented using Matlab at http://brain-imaging.waisman.wisc.edu/~chung/lb/ (Seo and Chung, 2011).

If we imagine starting from a spherical version of a brain hemisphere, the eigenmodes in that case are spherical harmonics Y_{lm}, an approximation that has been extensively used by Nunez, Jirsa, Seo, and others to represent brain activity, EEGs, and cortical shape (Nunez, 1974, 1995; Nunez and Srinivasan, 2006; Olver et al., 2010; Seo and Chung, 2011; Seo et al., 2010; Jirsa et al., 2002). Introducing the folding of the brain in stages will perturb those eigenmodes, and change their eigenvalues. Notably, even when the folding becomes large, the Courant Nodal Line Theorem (Gladwell and Zhu, 2002; Cheng, 1976; Courant and Hilbert, 1953; Kuttler and Sigillito, 1984) limits the complexity of the spatial structure of the resulting modes by restricting the number of separate regions that can have positive or negative sign to at most n for the nth eigenmode, taken in order with the lowest mode being counted as the first. This result depends on the ∇^2 form of the spatial

coupling in Eq. (12). Here we use η to denote $\lambda\mu$ in general when labeling modes and lm for the special case of a sphere. In the case of a sphere of radius R_s one has

$$k_{lm}^2 r_{ee}^2 = \left(\frac{r_{ee}}{R_s}\right)^2 l(l+1), \tag{33}$$

so the eigenvalues depend only on the angular momentum mode number $l=0,1,\ldots$ and are degenerate with respect to the azimuthal mode number $m=-l,\ldots,l$.

Equivalence of eigenmodes of activity and of connectivity matrices

Most commonly, functional connection matrices (fCMs) are calculated from covariances of BOLD signal fluctuations after removal of global components that are believed to be the most affected by artifacts (e.g., from heartbeats and breathing) (Murphy et al., 2009; He and Liu, 2012). Their eigenmodes are the same as those of brain activity, and of the direct effective CM (deCM), which measures only direct connections between points, including their strengths (Galán, 2008; Robinson, 2012; Robinson et al., 2014), and can sometimes be approximated by a scaled anatomical CM (aCM). Hence, eigenmodes obtained by solving NFT on the curved cortex can be directly compared with those seen in an eigenmode expansion of a deCM or fCM.

Expansions of eigenmodes and activity in spherical harmonics, and of activity in eigenmodes

It is often useful to express activity in terms of eigenmodes, which provide an extremely compact representation. Because these modes approach spherical harmonics in the limit of vanishing cortical folding, it is of interest to express them in terms of spherical harmonics. This is also of wider use because the properties of spherical harmonics are well understood and many insights can be gained from them Olver et al. (2010); Courant and Hilbert (1953); Nunez (1995); Nunez and Srinivasan (2006); Wingeier et al. (2001).

Any continuous bounded function f on a sphere can be expanded in terms of real-valued spherical harmonics, with

$$f(\theta,\varphi) = \sum_{lm} f_{lm} Y_{lm}(\theta,\varphi), \tag{34}$$

$$f_{lm} = \int f(\theta,\varphi) Y_{lm}(\theta,\varphi)\,\sin\theta\,d\theta\,d\varphi, \tag{35}$$

with l being a nonnegative integer and $-l\le m\le l$. In Eq. (34) the sum extends over all such l and m, while the integral in Eq. (35) extends over the whole sphere. Here

$$Y_{lm} = c_{lm} P_l^{|m|}(\cos\theta)\sin(|m|\varphi), \quad -l\le m\le -1, \tag{36}$$

$$= \frac{c_{lm}}{\sqrt{2}} P_l^0(\cos\theta), \quad m=0, \tag{37}$$

$$= c_{lm} P_l^{|m|}(\cos\theta)\cos(|m|\varphi), \quad 1\le m\le l, \tag{38}$$

$$c_{lm} = \left[\frac{2l+1}{2\pi}\frac{(l-|m|)!}{(l+|m|)!}\right]^{1/2}, \tag{39}$$

where the Y_{lm} are the real spherical harmonics and the P_l^m are the associated Legendre polynomials of order m (Olver et al., 2010).

We can expand brain activity $\phi(r,t)$ as in Eq. (34) to find

$$\phi(\boldsymbol{r},t) = \sum_{lm} \phi_{lm} Y_{lm}(\theta,\varphi)\exp(-i\omega_{lm}t), \tag{40}$$

where $\boldsymbol{r}=\boldsymbol{r}(\theta,\varphi)$ as in the mapping discussed in Inflation of cortical surface into a sphere for visualization and to investigate intermediate shapes section and ω_{lm} is the solution of Eq. (31) with η denoting lm. By analogy, we can also expand activity in terms of orthonormal eigenmodes $y_{\lambda\mu}$, where λ and μ generalize l and m, respectively. This yields the most compact expression

$$\phi(\boldsymbol{r},t) = \sum_{\lambda\mu} \phi_{\lambda\mu} y_{\lambda\mu}(\boldsymbol{r}) e^{-i\omega_{\lambda\mu}t}, \tag{41}$$

$$\phi_{\lambda\mu} = \int \phi(\boldsymbol{r},0) y_{\lambda\mu}(\boldsymbol{r})\,dS, \tag{42}$$

where the sums are over all the l and m, dS denotes an element of cortical surface area, and the integral is over the whole surface.

Once we have calculated the eigenmodes $y_{\lambda\mu}$ of corticothalamic activity on a cortical hemisphere we can map these onto a sphere via the mapping of the Inflation of cortical surface into a sphere for visualization and to investigate intermediate shapes section to obtain the projection $y_{\lambda\mu}(\theta,\varphi)$. The eigenmodes increasingly resemble spherical harmonics for surfaces that are less folded, becoming identical in the spherical limit. Hence, it is important to quantify the similarity and to determine how much the eigenmode structure is changed by folding. We do this by expanding each eigenmode's representation on the sphere in terms of spherical harmonics, as in Eq. (34) with $f(\theta,\varphi)=y_{\lambda\mu}(\theta,\varphi)$, which gives expansion coefficients $y_{\lambda\mu lm}$ of the $\lambda\mu$ eigenmode in terms of the lm spherical harmonics.

One complication in the analysis of eigenmodes is that even the slightest cortical folding breaks the rotational symmetry of the sphere. So the eigenmodes of a cortical hemisphere will usually be rotated relative to the standard spherical harmonics in any initially assumed coordinate system of the sphere to which they are mapped. Because l is an angular momentum mode number, rotation in the exactly spherical case affects only the m values, implying $y_{\lambda\mu lm}=0$ for $\lambda\neq l$ in this limit. So it is of most interest to determine how much power in the expansion also stays in the same l for eigenmodes on a convoluted brain hemisphere. Hence, we calculate

$$p_{\lambda\mu l} = \sum_m |y_{\lambda\mu lm}|^2, \tag{43}$$

$$p_{\lambda l} = \sum_{\mu m} |y_{\lambda\mu lm}|^2 = \sum_{\mu} p_{\lambda\mu l}. \tag{44}$$

The first of these quantities quantifies the total contribution to the $\lambda\mu$ eigenmode from the lm spherical harmonics for fixed l and all m. The second quantity is the total contribution to the eigenmodes $\lambda\mu$, summed over μ, from eigenmodes lm summed over m; i.e., this tells us how much of the set of eigenmodes at a given λ is contributed by the set of spherical harmonics of order l.

Determination of connectivity matrix eigenvectors and eigenvectors

The eigenvalues a_j and corresponding eigenvectors $\boldsymbol{u}_j$ of an experimental deCM $\mathbf{A}$ are obtained by solving the eigenvalue equation

$$\boldsymbol{A}\boldsymbol{u}_j = a_j \boldsymbol{u}_j, \tag{45}$$

where the $\boldsymbol{u}_j$ are the eigenvectors and the a_j are the corresponding eigenvalues. Numerically, we use the standard Matlab eig routine to solve this equation via QR factorization. The eigenvalues and corresponding eigenvectors are then reordered in descending order to place the dominant eigenmodes first.

Connectivity matrix and cortical surface data

The present work draws on two sources of experimental data. First, we employ a standard published digitized hemispheric surface on

which to solve our NFT numerically for comparison with experimental data (Fischl et al., 1999). This is the surface provided standard with Freesurfer, which uses 163,842 vertexes to define 327,680 faces.

Second, we use a published experimental anatomical CM (aCM) to approximate the deCM in the low-frequency limit, since its elements are estimates of the strengths of structural connections between cortical regions. The matrix used is based on group-averaged data from the NKI-Rockland sample (Nooner et al., 2012), obtained in processed form from the USC Multimodal Connectivity Database (http://umcd.humanconnectomeproject.org). A detailed description of the analysis procedures was given by Brown et al. (2012). In brief, these authors derived a 188-node parcellation of cortical, subcortical, and cerebellar regions from "resting state" fMRI data using the spectral clustering method of Craddock et al. (2013). This yielded group-level RoIs with volumes ranging from 84 to 540 mm^3. These RoIs were then projected into diffusion space for each of their 60 subjects and used to construct aCMs in which the values of elements were the numbers of deterministic tractography streamlines passing through at least one voxel in both RoIs. The 60 individual CMs were then averaged elementwise to give the group-average aCM whose intracortical parts are used in the present work, giving a 166×166 matrix overall, with half the nodes per hemisphere.

Results

In this section we first use the NFT of the Methods section to calculate the eigenmodes on a cortical hemisphere and study their morphology, dependence on folding, and relationship with the spherical harmonics. This is followed by comparisons with eigenfunctions of experimental deCMs. The results are then extended to the bihemispheric case via a generalized NFT analysis and use of bihemispheric deCMs in the experimental case. Finally, the eigenvalues are used to constrain the strength of connectivity between hemispheres.

Unihemispheric eigenmodes from NFT

To calculate the eigenmodes of a single brain hemisphere, we first write the dispersion relation (31) [with Eq. (27)] in the form

$$(1-i\omega_\eta/\gamma_{ee})^2 + k_\eta^2 r_{ee}^2 - J = 0, \tag{46}$$

$$J = J_{ee}/(1-J_{ei}-J_{ctc}). \tag{47}$$

In the case of a spherical approximation to each brain hemisphere, a value of $R_s \approx 100$ mm for one cortical hemisphere follows from the requirement that the total area of the two hemispheres match the observed 0.25 m^2 of the typical whole adult cortex Kandel et al. (2000). This implies that the coefficient of $l(l+1)$ on the right of Eq. (33) is $(r_{ee}/R_s)^2 \approx 0.73$ since $r_{ee} \approx 85$ mm (Robinson, 2005; Robinson et al., 2004).

Once k_η^2 has been found, Eq. (46) determines the stability and frequency of eigenmodes. Modes must have $Im\omega < 0$ to be stable and the stability boundary of the system occurs where $Im\omega = 0$. For example, if J is real in Eq. (46), which is a reasonable approximation for $\omega/2\pi$ much less than the reciprocals of characteristic time constants in the system (i.e., for $\omega \ll 60\ s^{-1}$), then

$$\frac{\omega_\eta}{\gamma_{ee}} = -i \pm \left[k_\eta^2 r_{ee}^2 - J\right]^{1/2}. \tag{48}$$

Unlike Eq. (46) at general ω, Eq. (48) is not transcendental in ω and can be readily solved in the low-frequency case. It implies that the least stable low-frequency eigenmode occurs at $k_\eta = 0$ and is stable only for $J \le 1$ (Robinson et al., 1997). The boundary $J = 1$ corresponds to a critical point at which the system has unit gain and a saddle-node bifurcation occurs (Robinson et al., 1997, 2002; Breakspear et al., 2006). This has been associated with a critical $1/f$ spectrum in EEG (Robinson et al., 1997, 2002); related power-law behavior of critical-state neural avalanches was later found in cell cultures (Beggs and Plenz, 2003).

Since $J < 1$ is required for stability at low ω, Eq. (33) and analogs for the folded cortex imply that there are at most only a few eigenmodes for which the contents of the square brackets in Eq. (48) are negative, giving non-oscillatory solutions. Other eigenmodes are intrinsically oscillatory, but damped, with an angular frequency given by the square root in Eq. (48) for real J. From Eq. (48) we then find that the damping rate of the least damped eigenmode at $k_\mu = 0$ is $\gamma_{ee}(1-\sqrt{J})$ (Robinson et al., 1997). For typical waking-state parameters $\gamma_{ee} = 116\ s^{-1}$ and $J \approx 0.88$ (Robinson et al., 1997, 2014, 2004; Robinson, 2012) this gives a damping rate of about $7\ s^{-1}$ for the least damped eigenmode, hence implying that the spectral width of this eigenmode is of the same order — i.e., ~1 Hz in cyclic frequency.

Experimental determination of fCMs overwhelmingly relies on low-frequency BOLD signals, so we are mostly concerned with this case in the present work and correspondingly assume that J is real henceforth. However, as an aside, we note that if one allows for the frequency dependence of J, a more complex stability zone is obtained in the parameter space of the model, beyond which nonlinear phenomena such as seizures occur via bifurcations (Robinson et al., 2002; Breakspear et al., 2006; Roberts and Robinson, 2008). One approximation that could be made is

$$J(\omega) \approx J_0(\omega) + J_1(\omega)e^{i\omega t_0}, \tag{49}$$

where J_0 and J_1 are functions with low-pass characteristics, and the term $e^{i\omega t_0}$ incorporates the effects of the corticothalamic loop feedback with a total delay t_0. The two terms on the right of Eq. (49) change their relative phases as ω increases, giving rise to resonances at the alpha and beta frequencies ($\omega \approx 2\pi/t_0$ and $4\pi/t_0$, respectively) in waking, for example (Robinson et al., 2002; Breakspear et al., 2006). These will be relevant to higher-frequency connectivities measured via EEG and MEG, for example.

Fig. 3(a) shows the eigenvalues of the first 16 eigenmodes obtained by solving Eq. (30) on curved surfaces ranging from the sphere of radius $R_s = 100$ mm to a fully folded left cortical hemisphere from Freesurfer (see the Connectivity matrix and cortical surface data section) as a function of the degree of folding ρ (defined here to be the fraction of the total unfolding-algorithm steps remaining in going from folded cortex to sphere) at various stages of the inflation algorithm described in the Inflation of cortical surface into a sphere for visualization and to investigate intermediate shapes section. In the spherical case at left, all eigenvalues with the same l are equal, with values given by Eq. (33). As folding increases, the averages of the eigenvalues over μ at each λ are almost constant (exactly so for $\lambda = 0$), while the eigenvalues for individual μ exhibit folding-induced splitting, which increases for higher λ. There is no eigenvalue crossing, as is particularly evident near the rightmost ends of the two lowest $\lambda = 3$ eigenvalues, which approach one another very closely, then reverse direction to avoid crossing. Very similar results are obtained if the right hemisphere is used, as seen in Fig. 3(b), where the behavior of the two lowest $\lambda = 3$ eigenvalues is clearer on the scale of the figure because they do not approach one another quite so closely. Note that the eigenvalue structure differs most at the right ends of the curves, corresponding to full folding, although the differences are slight even there.

Fig. 4 shows the spatial structure of the first 9 NFT eigenmodes on the fully folded left brain hemisphere used in Fig. 3(a), with the eigenmodes labeled by λ and μ; the latter index has been arbitrarily assigned here in order of energy from $\mu = -\lambda$ to $\mu = \lambda$ in each case, but symmetry breaking implies that these remain superpositions of multiple m eigenmodes. This also applies if the cortex is inflated toward a sphere, except for three special orientations of the spherical coordinates. The $\lambda = 0$ eigenmode is approximately uniform, with no nodal lines. The three $\lambda = 1$ eigenmodes each have enhanced activity in one half of the

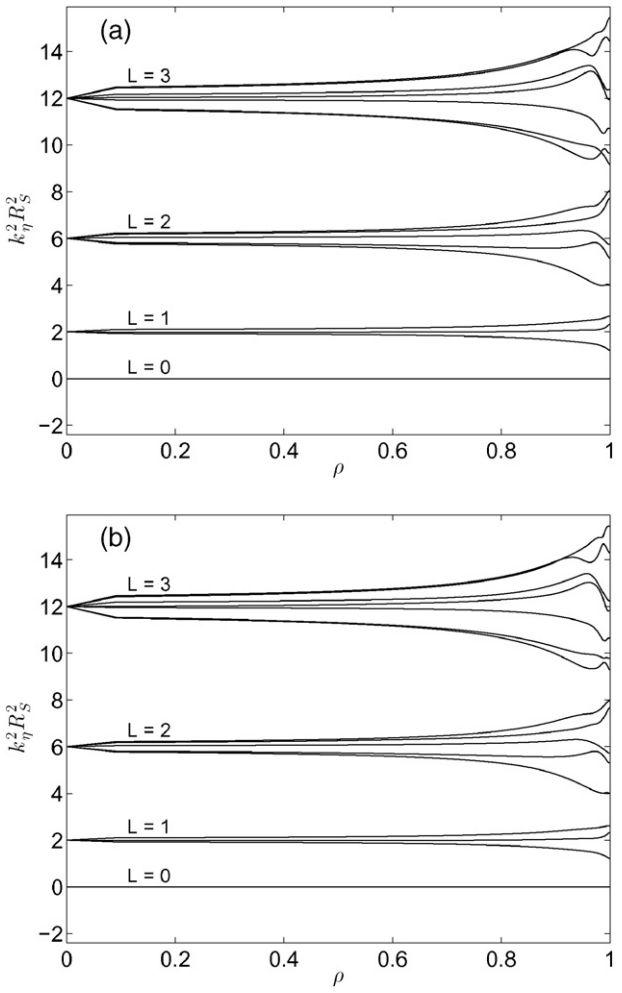


Fig. 3. Eigenvalue spectrum of the first 16 eigenmodes of brain hemispheres obtained by solving Eq. (30) on a curved cortex. The degree of folding, ρ, starts at a spherical brain hemisphere at $\rho = 0$ and reaches the fully folded brain hemisphere described in the Connectivity matrix and cortical surface data section at $\rho = 1$. (a) Left hemisphere. (b) Right hemisphere.

brain hemisphere and reduced activity in the other half, with one nodal line, as in the $l = 1$ spherical harmonics. The $\lambda = 2$ eigenmodes have very similar structure to the $l = 2$ spherical harmonics, aside from perturbations replacing nodal-line crossings with near-approaches of "reconnected" nodal lines that turn back just before intersecting — precise intersection occurs in the limit as the cortex becomes spherical. We have been careful to verify that there is no eigenmode crossing — a high resolution plot shows that the curves turn back smoothly without approaching close enough for there to be misidentification of modes due to numerical inaccuracies. Moreover, the spatial structure of the lower mode is the same on both sides of the near-approach, and different from that of the upper mode, whose structure is also the same on each side of this point; if there had been a crossover, the two structures would have been exchanged.

We find that the eigenmodes reduce to the corresponding spherical harmonics for a spherical geometry. The eigenmodes are also found to be orthogonal and we normalize them so they are orthonormal. Equivalent results to Fig. 4 are seen for the right hemisphere in Fig. 5, demonstrating that the eigenmodes are almost identical, aside from being reflected across the brain's midplane.

The assignment of λ values to the NFT eigenmodes in Figs. 4 and 5 is further supported by examination of the decomposition of these eigenmodes into spherical harmonics, as discussed in the Expansions of eigenmodes and activity in spherical harmonics, and of activity in eigenmodes section. We see from Figs. 6 and 7, respectively, that all eigenmodes are dominated by spherical harmonic components with $l = \lambda$, despite the heavy folding of the actual brain hemisphere. The contribution from other values of l increases at higher λ. We note that each $\lambda\mu$ eigenmode is predominantly a linear combination of all the Y_{lm} with $l = \lambda$, not just a single m. This parallels the fact that, even for a sphere, a given Y_{lm} is a linear combination of all the spherical harmonics of the same l in a rotated coordinate system.

Eigenmodes of a unihemispheric experimental deCM

We next determine the eigenvectors of the experimental left-hemisphere anatomical CM described in the Connectivity matrix and cortical surface data section, which we treat as providing an estimate of the deCM $\mathcal{N}^{(0)}$ in the static ($\omega = 0$) limit (Robinson, 2012; Robinson et al., 2014). This matrix is the upper left submatrix of the full aCM (i.e., one quarter of the whole matrix), embodying connections between left-hemisphere nodes. Use of this matrix enables us to compare its eigenmodes with those of NFT without the complicating effects of interhemispheric connections, whose influences cannot be avoided when dealing with functional CMs, for example.

Solving for the eigenvectors of this matrix yields the eigenmodes seen in Fig. 8. Despite the interference of the unsampled region around the corpus callosum, shown in gray in Fig. 8, and the coarse resolution of the experimental results (which means that nodal lines are not well resolved) the correspondence between these eigenmodes and those from NFT in Fig. 4 remains close, including numbers and orientations of nodal lines, especially for $\lambda = 0, 1$. Experimental modes appear to mix some characteristics of NFT modes at the same λ, especially for $\lambda = 2$, an effect that is likely to be due to the effects of the region around the corpus callosum. A quantitative analysis of modal correspondences is carried out below.

Fig. 9 shows the right-hemisphere eigenmodes for the same experimental dataset as in Fig. 8, using the lower-right submatrix of the full CM. Comparisons with the NFT eigenmodes in Fig. 5 and with the left-hemisphere results again show parallels in structure of eigenmodes and locations of nodal lines for $\lambda = 0, 1$, with more mode mixing at $\lambda = 2$.

The similarities between the NFT eigenmodes and the experimental deCM eigenmodes are quantified objectively in Figs. 10 and 11, which show the spherical-harmonic decomposition of the experimental eigenmodes, analogously to Figs. 6 and 7. We see that the eigenmodes with $\lambda = 0, 1$ and the middle three $\lambda = 2$ eigenmodes are again dominated by spherical harmonic components with $l = \lambda$, despite the heavy folding of the cortex. For $\lambda = 2$ there is more admixture of $l = 1$ for the lowest eigenmode ($\mu = -2$), falling rapidly with increasing μ; likewise there is a larger admixture of $l = 3$ in the $\mu = 2$ eigenmode, with the opposite trend vs. μ. These trends are as expected for first-order departures from $\lambda = l$ dominance — eigenmodes nearest to neighboring l eigenmodes will have the greatest mixing of characteristics.

The greater admixture of eigenmodes with $l \neq \lambda$ in the experimental deCM vs. the NFT ones warrants further examination. We argued above that this is likely to be due to the effects of the gray regions around the corpus callosum in Figs. 8 and 9 where there are no observations — a gap that acts as an additional perturbation relative to spherical eigenmodes. Notably, in Fourier analysis, and orthogonal expansions in general, the presence of a sharp edge like this in a function generates high-order coefficients in its representation in terms of orthogonal functions, typically with a slowly decreasing spectrum. Indeed, examination of higher-l terms in extended histograms of the type shown in Figs. 10 and 11 reveals exactly this behavior as l increases, with significant energy in eigenmodes with internodal distances down to the same order as the size of the missing region. We have further tested this point by zeroing out a similar sized region in the NFT eigenmodes before evaluating their spherical-harmonic expansion. We find similar effects to those seen in Figs. 10 and 11, with the same pattern of admixture of neighboring l

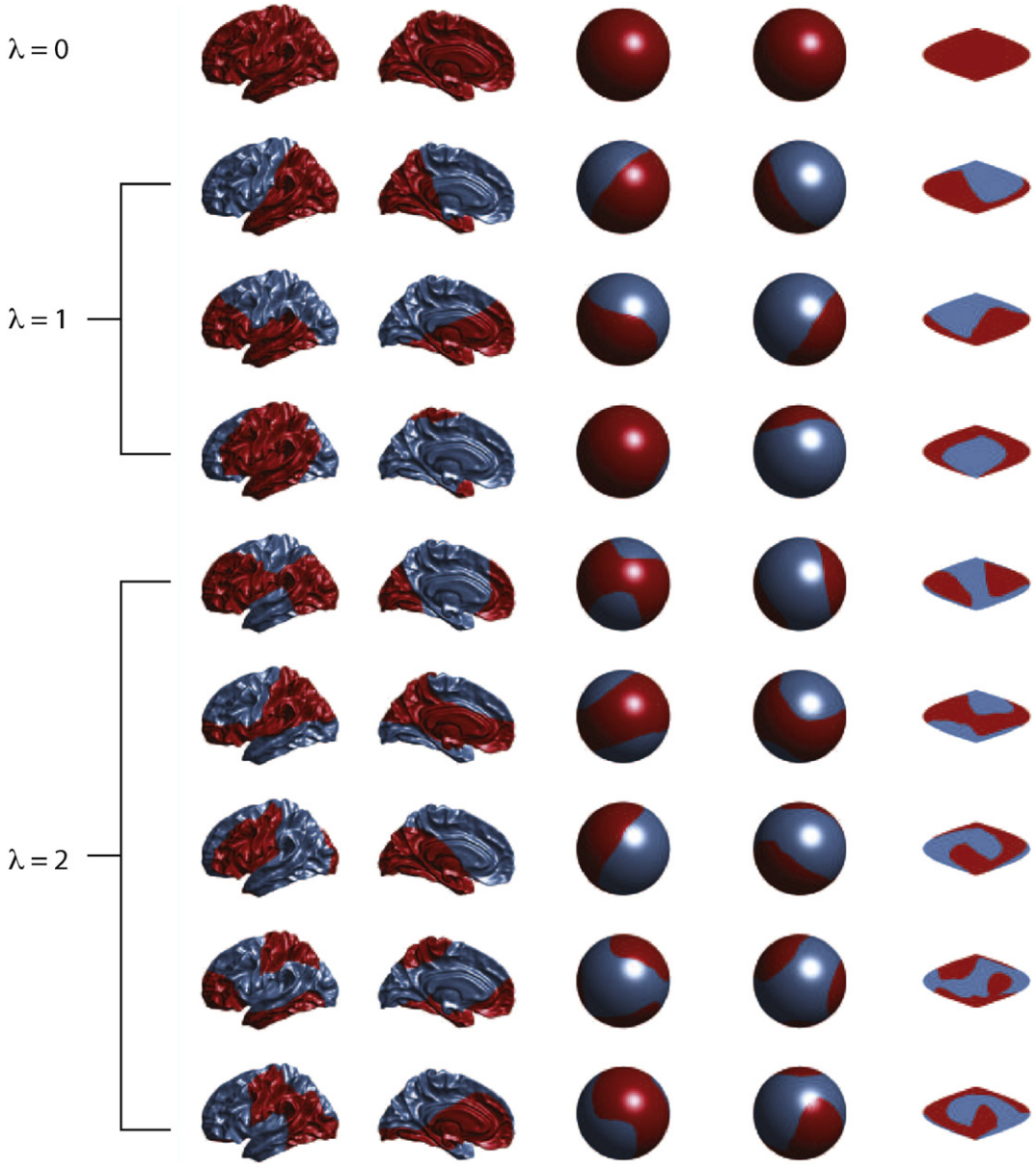


Fig. 4. First 9 NFT eigenmodes of the left brain hemisphere corresponding to Fig. 3(a), starting with the lowest eigenmode at top, ordered by eigenvalue and labeled with mode numbers $\lambda\mu$, as shown. Red indicates positive regions and blue negative ones. In each row the first and second columns show two views on the cortical hemisphere, the third and fourth columns show these inflated onto a sphere, and the fifth column shows the flattened projection defined in Flattening of cortical surface onto a plane for visualization section.

values, and significant energy at high l, but with lower amplitudes of the admixed eigenmodes. Conversely, we can smooth the edges of the gray regions in Figs. 8 and 9 by multiplying neighboring eigenfunction values by functions that decrease smoothly to zero (rather than having sharp edges). This reduces the amplitudes of the admixed eigenmodes, yielding histograms more like those in Figs. 6 and 7. Another possible source of eigenmode coupling is the relatively coarse-grained discretization of the NKI data, which have only around 80 nodes per hemisphere (see the Connectivity matrix and cortical surface data section). This gives a typical spacing of roughly 4 cm, which is non-negligible relative to the characteristic ~15 cm internodal distance of $\lambda = 2$ eigenmodes. A further possible source of coupling is that we have calculated our NFT eigenmodes here on the basis of gray matter distances, whereas Henderson and Robinson (2013, 2014) showed that white matter distances are a better measure. Overall, we have thus only partly resolved this issue and a systematic treatment of these effects is needed to obtain a complete solution, but this is beyond the scope of the present paper.

Bihemispheric eigenmodes from NFT

To extend the analysis of the Corticothalamic NFT of one hemisphere section to the bihemispheric case, we generalize the various quantities by adding the superscript H, with $H = R$ and $H = L$ denoting right and left hemispheres, respectively. Fibers also cross between homologous points in opposite cortical hemispheres, so we also add interhemispheric couplings I_{ee}^{RL}, I_{ie}^{RL}, I_{ee}^{LR}, and I_{ie}^{LR}, which are analogous to the J_{ab}; although generally thought to be excitatory, we leave their nature open in this section and determine it in the Eigenmodes of a bihemispheric experimental deCM section. Here the superscript LR denotes signals to the left hemisphere from the right, and RL the reverse. In the random

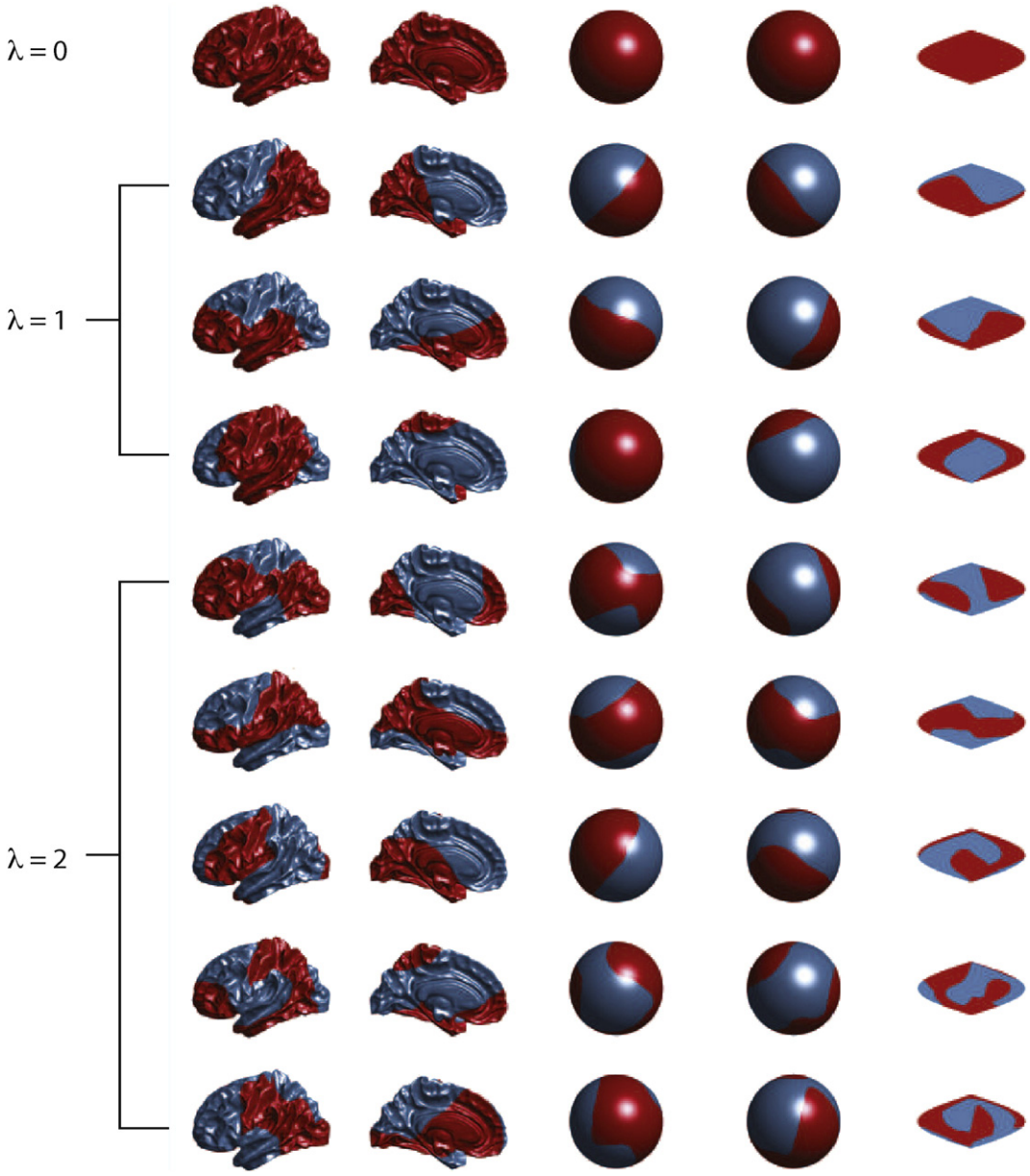


Fig. 5. First 9 NFT eigenmodes of the right brain hemisphere corresponding to Fig. 3(b), starting with the lowest eigenmode at top, ordered by eigenvalue and labeled with mode numbers λμ as shown. Red indicates positive regions and blue negative ones. In each row the first and second columns show two views on the cortical hemisphere, the third and fourth columns show these inflated onto a sphere, and the fifth column shows the flattened projection defined in the Flattening of cortical surface onto a plane for visualization section.

connectivity approximation mentioned in the Corticothalamic NFT of one hemisphere section, $I_{ee}^{LR} = I_{ie}^{LR}$ and $I_{ee}^{RL} = I_{ie}^{RL}$. Moreover, we will assume that the feedbacks are symmetric between hemispheres, which is a good first-order approximation, and simply denote the couplings by I_{ee}. By analogy with the intrahemispheric J_{ab}, I_{ee} is of the form

$$I_{ee}(\omega) = G_l L_l(\omega) e^{i\omega t_l/2}, \tag{50}$$

where G_l is the gain of interhemispheric connections, L_l is the relevant synaptodendritic lowpass response function, and $t_l/2$ is the typical transcallosal axonal propagation time.

Estimated values of t_l cover a broad range from a few ms to a few tens of ms because there is a range of axonal velocities involved (Roberts and Robinson, 2008; Bojak and Liley, 2010). At the low frequencies of fMRI signals, these delays can be ignored, but they need to be retained in higher-frequency EEG applications. The gain G_l must be non-negligible relative to G_{ee}, based on: (i) the fact that the detection of regional interconnections via transcallosal fibers is common in anatomical, structural, and functional imaging studies [although some methods such as diffusion weighted MRI systematically underestimate the numbers of such connections (Deco et al., 2014; Robinson et al., 2014)], (ii) the likelihood of evolutionary elimination of these fibers if they were nonfunctional, and (iii) the occurrence of significant functional abnormalities in subjects born without a corpus callosum (about 0.1% of the population) or who have had it surgically severed (Berg and Vickrey, 1997).

Using the above notation, Eq. (20) generalizes to

$$\left[\mathcal{D}_{ee} - \frac{J_{ee}}{1-J_{ei}-J_{ctc}}\right]\phi_{ee}^{R,L} - \frac{I_{ee}}{1-J_{ei}-J_{ctc}}\phi_{ee}^{L,R} = \frac{J_{es}J_{sn}}{(1-J_{sr}J_{rs})(1-J_{ei}-J_{ctc})}\phi_{sn}^{R,L}, \tag{51}$$

with Eqs. (21) and (22), where we have omitted the superscripts from

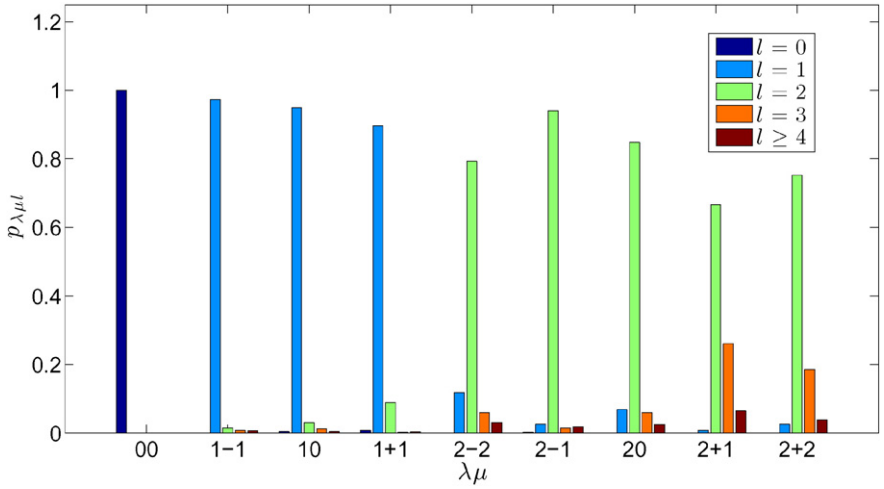


Fig. 6. Spherical harmonic decompositions of the first 9 left-hemisphere NFT eigenmodes from Fig. 4 as measured by the coefficients $p_{\lambda\mu l}$ in Eq. (44), with μ values arbitrarily assigned in order of energy. For each eigenmode $\lambda\mu$, the histogram shows the coefficients for $l = 0, \ldots, 3, 4^+$. The total of the bars in each histogram is unity because the spherical harmonics are orthonormal.

the J_{ab} under the approximation that the two hemispheres have the same physiological parameters, including those relating to morphology, but do not necessarily exhibit the same activity. Eq. (51) is of the same general form as Eqs. (4) and (5) in Neural field theory of brain eigenmodes section. In our more compact notation, this becomes

$$[\mathcal{D} - J]\phi_{ee}^{R,L} - I\phi_{ee}^{L,R} = C\phi_{sn}^{R,L}, \tag{52}$$

$$I = I_{ee}/(1 - J_{ei} - J_{ctc}). \tag{53}$$

In seeking eigenmodes of the coupled system, we set the right side of Eq. (51) to zero. This leads to two solutions

$$[\mathcal{D} - J - \varepsilon\mathrm{I}]\phi_{ee}^{\varepsilon} = 0, \tag{54}$$

with $\varepsilon = \pm 1$ and $\phi_{ee}^{L} = \varepsilon\phi_{ee}^{R}$; these bihemispheric whole-system solutions have the advantage of being mutually orthogonal. At low frequency, symmetric solutions have $\phi_{ee}^{L} = \phi_{ee}^{R}$ and the effects of I add in phase with those of J. Low-frequency antisymmetric solutions have $\phi_{ee}^{L} = -\phi_{ee}^{R}$ with I and J out of phase. These results are exactly as in standard theory of coupled oscillators, and we stress that the coupling in Eq. (54) causes no change in the spatial structure of the solutions within a given hemisphere — eigenmodes simply have an added symmetry or antisymmetry between hemispheres, with the original spatial modal structure within each. We note that symmetry of solutions between homologous regions in the coordinates in Fig. 1 corresponds to mirror-symmetry across the saggital midplane of the brain in a global coordinate system, so points with the same $\boldsymbol{r}$ have the same activity.

By analogy with Eq. (26) we now find in a given hemisphere H

$$\left[(1 - i\omega/\gamma_{ee})^2 - J - \varepsilon\mathrm{I}\right]\phi_{ee}^{\varepsilon\mathrm{H}} = r_{ee}^2\nabla^2\phi_{ee}^{\varepsilon\mathrm{H}}, \tag{55}$$

where the superscript εH indicates the symmetry of the solution between hemispheres and the hemisphere being examined. The separation of variables then proceeds as in the Solution of the Helmholtz equation on cortical hemisphere to obtain NFT eigenmodes section. This leads to the same single-hemisphere spatial eigenfunctions as before, extended by symmetry or antisymmetry to the whole brain. Similarly to Eq. (22), ω_η satisfies

$$(1 - i\omega_\eta/\gamma_{ee})^2 + k_\eta^2 r_{ee}^2 - J - \varepsilon\mathrm{I} = 0, \tag{56}$$

where η now denotes the set of mode numbers $\lambda\mu\varepsilon$ and Eq. (56) has the same form as Eq. (46) except that J has been replaced by $J + \varepsilon\mathrm{I}$.

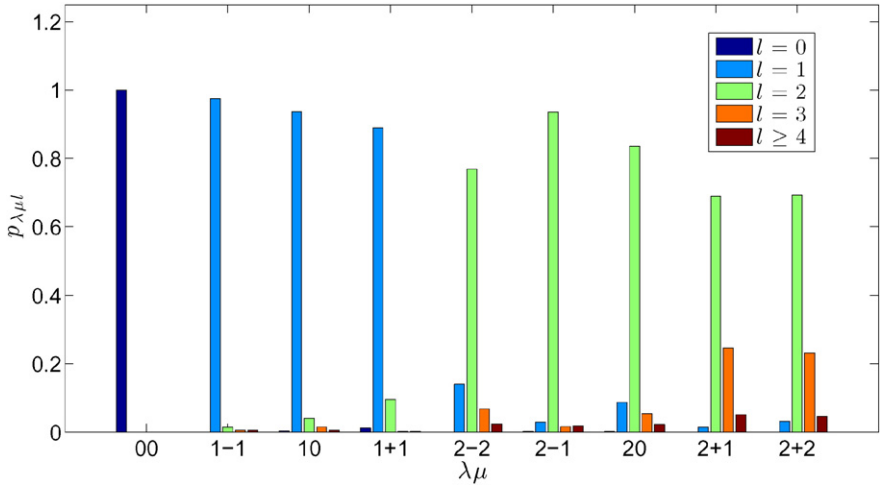


Fig. 7. Spherical harmonic decompositions of the first 9 right-hemisphere NFT eigenmodes from Fig. 5 as measured by the coefficients $p_{\lambda\mu l}$ in Eq. (44), with μ values arbitrarily assigned in order of energy. For each eigenmode $\lambda\mu$, the histogram shows the coefficients for $l = 0, \ldots, 3, 4^+$. The total of the bars in each histogram is unity because the spherical harmonics are orthonormal.

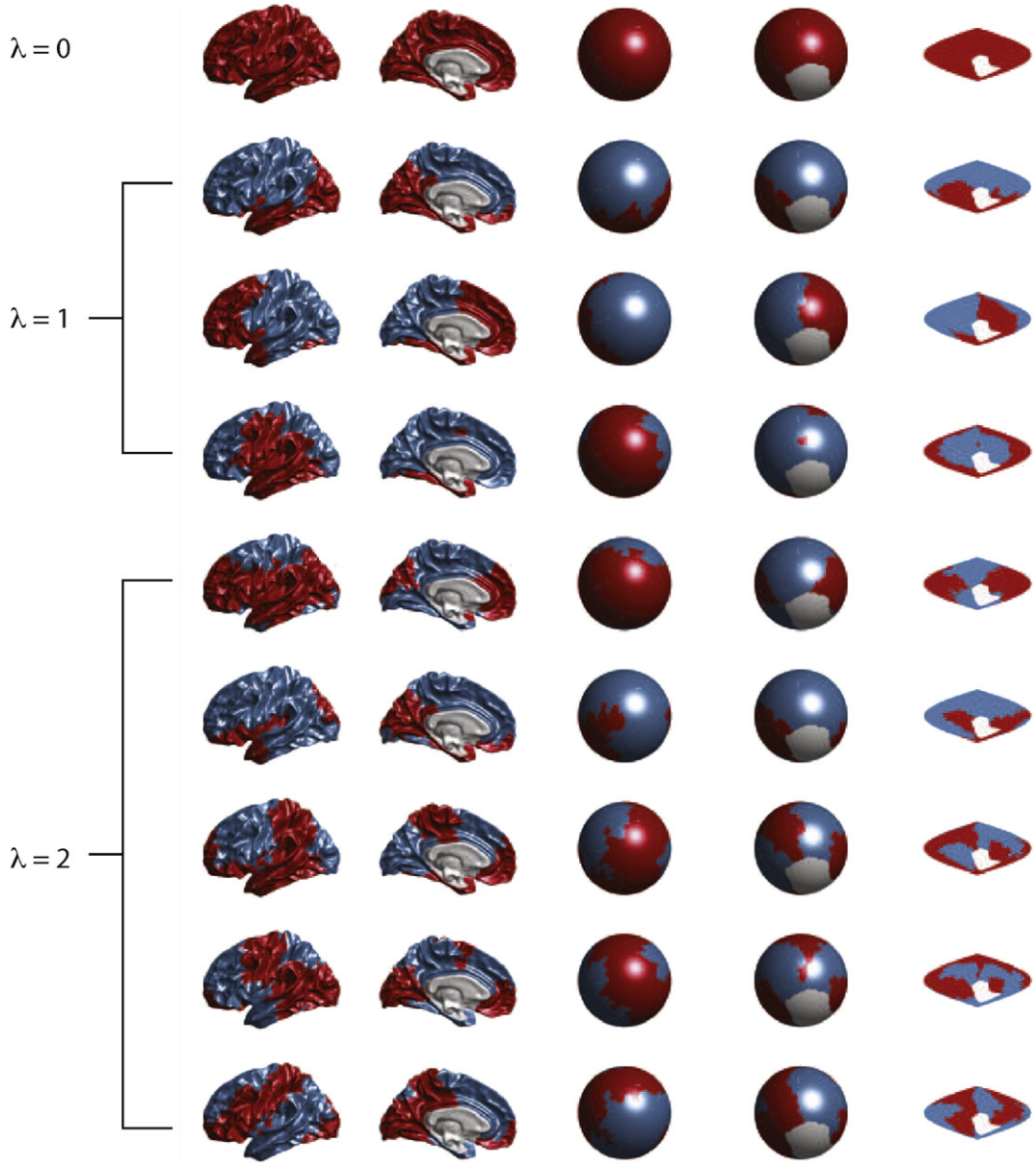


Fig. 8. First 9 eigenmodes of the experimental deCM of the left hemisphere for the data in the Connectivity matrix and cortical surface data section, starting with the lowest eigenmode at top, ordered by eigenvalue and labeled with mode numbers λμ as shown. Red indicates positive regions and blue negative ones. In each row the first and second columns show two views on the cortical hemisphere, the third and fourth columns show these inflated onto a sphere, and the fifth column shows the flattened projection defined in the Flattening of cortical surface onto a plane for visualization section.

In parallel to the analysis of Eq. (46), the quantity

$$E_\eta^2 = k_\eta^2 r_{ee}^2 - J - \varepsilon \mathrm{I}, \tag{57}$$

expresses both eigenmode stability, which requires $E_\eta^2 > -1$, and eigenmode frequency, with $\mathrm{Re}\omega_\eta = \gamma_{ee} E_\eta$ for real positive $E_\eta^2 > 0$. At $\omega = 0$, J and I are real, so low-ω stability of the bihemispheric brain requires $|J + \varepsilon \mathrm{I}| < 1$. For $k_\eta = 0$ the eigenmodes are damped and nonoscillating, whereas for $k_\eta^2 r_{ee}^2 \gg 1$ one finds damped propagating waves with $\omega_\eta \approx \pm k_\eta \gamma_{ee} r_{ee} - i\gamma_{ee}$; for $R_s = 100$ mm all eigenmodes with nonzero k_η are expected to be oscillatory. If the transcallosal connections are excitatory, I is positive and symmetric eigenmodes have lower energy and are less stable than their antisymmetric counterparts for the same k_η. Again, this is a typical result from coupled-oscillator theory. Here, it implies that symmetric eigenmodes of activity will typically have higher amplitudes than antisymmetric ones if excitation is random. Moreover, cutting the corpus callosum, or otherwise suppressing signaling through it, will tend to stabilize the brain, consistent with results on split-brain epileptic patients, who showed stabilization of seizure activity after callosectomy (Berg and Vickrey, 1997).

If J and I can be approximated as real in Eq. (55) — e.g., at the low frequencies relevant to fMRI — we can analyze the above features using the diagram shown in Fig. 12. This figure shows $E_\eta^2 + J$ vs. I for positive I, using the averages of the experimental left- and right-hemisphere deCM eigenvalues for the intercepts at $I = 0$. Because $J + I < 1$, the region below -1 on the vertical axis corresponds to instability and is unphysical, thereby implying $I < 1$ because every eigenmode must be stable for the brain to be stable overall. In fact, the region below $J - 1$ is unstable, but we have not determined J at this point in the analysis (see the Eigenmodes of a bihemispheric experimental deCM section).

Fig. 12 shows how each unihemispheric eigenmode is split into a pair by interhemispheric interactions; one member of each pair is symmetric between hemispheres and one antisymmetric, and their separation grows as I increases, with the symmetric one having lower E_η^2 for

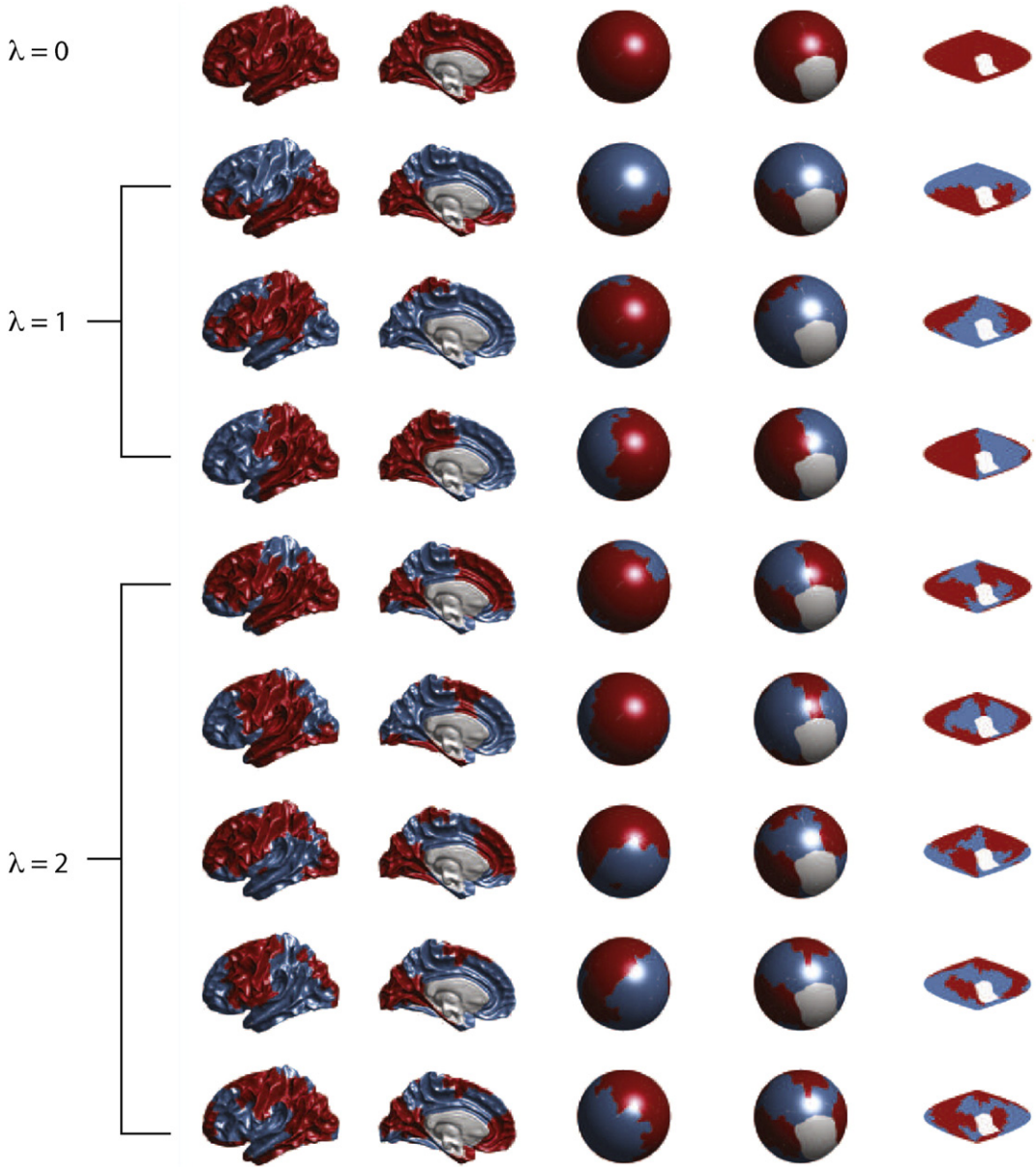


Fig. 9. First 9 eigenmodes of the experimental deCM of the right hemisphere for the data in the Connectivity matrix and cortical surface data section, starting with the lowest eigenmode at top, ordered by eigenvalue and labeled with mode numbers λμ as shown. Red indicates positive regions and blue negative ones. In each row the first and second columns show two views on the cortical hemisphere, the third and fourth columns show these inflated onto a sphere, and the fifth column shows the flattened projection defined in the Flattening of cortical surface onto a plane for visualization section.

positive I, and a separation

$$\Delta E_\eta^2 = 2I. \tag{58}$$

In the bihemispheric case eigenmodes can cross one another if the splitting is large enough, as seen in Fig. 12. The preponderance of symmetric eigenmodes at the lowest (hence least stable) E_η^2 immediately explains the dominance of symmetric brain activity. This dominance will be further reinforced to the extent that many inputs to the brain are symmetric, or nearly so — e.g., many visual and auditory stimuli, which reach homologous regions in both hemispheres.

From Eq. (56) we obtain the result for the deCM eigenvalue $\lambda_u^{(0)}$ of the lowest-energy eigenmode

$$\lambda_u^{(0)} = J + |I|, \tag{59}$$

with the lowest eigenmode's ε having the same sign as I.

If we consider the two bihemispheric eigenmodes (labeled 00ε) that emerge from the lowest-energy unihemispheric eigenmode, and denote their eigenvalues by $\lambda_d^{(0)}$ and $\lambda_u^{(0)}$, with $\lambda_u^{(0)}$ being the larger of the two (i.e., the eigenvalue of the lowest eigenmode), we find

$$I = \frac{\varepsilon}{2}\left[\lambda_u^{(0)} - \lambda_d^{(0)}\right], \tag{60}$$

$$J = \frac{1}{2}\left[\lambda_u^{(0)} + \lambda_d^{(0)}\right], \tag{61}$$

where ε is the parity of the lowest eigenmode.

Eigenmodes of a bihemispheric experimental deCM

In Fig. 13 we show the eigenmodes obtained from the bihemispheric deCM described in the Connectivity matrix and cortical surface data section, where we noted that this deCM most likely misses some

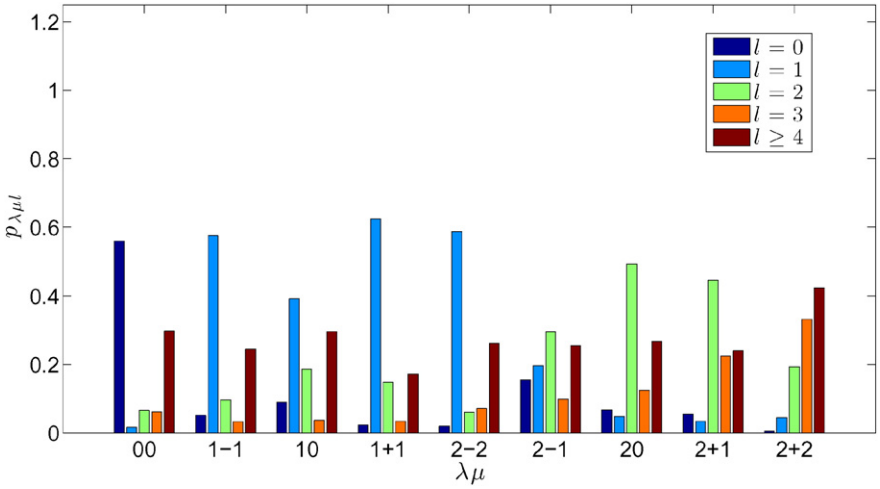


Fig. 10. Spherical harmonic decompositions of the first 9 left-hemisphere experimental deCM eigenmodes from Fig. 8 as measured by the coefficients $p_{\lambda\mu l}$ in Eq. (44), with μ values arbitrarily assigned in order of energy. For each eigenmode $\lambda\mu$, the histogram shows the coefficients for $l = 0, \ldots, 3, 4^+$. The total of each histogram is unity because the spherical harmonics are orthonormal.

interhemispheric connections. Because the deCM is estimated here from the static aCM, it also omits information on the time delays that would be required to analyze wave dispersion. Hence, the eigenvalues of nonzero-frequency eigenmodes, and their ordering, are not the same as those of oscillatory eigenmodes in NFT and the physical brain. However, they can be compared in the case $k_\eta = 0$; i.e., for the nonoscillatory 00ε eigenmodes, where Eqs. (60) and (61) remain valid. The spatial structure of eigenmodes does not depend on these temporal aspects and deCM data can thus be used to investigate spatial features.

The lowest eigenmode in Fig. 13 is symmetric between hemispheres, which immediately yields $I > 0$, thereby implying excitatory connections between hemispheres, in accord with physiological expectations, based on these connections being supported by white matter fibers that project from cortical pyramidal neurons. The lowest eigenmode is uniform within each hemisphere, as expected from Fig. 12 and is thus the 00+ eigenmode. The second eigenmode is almost antisymmetric between hemispheres (we do not expect experimental results to be exactly antisymmetric because the hemispheres are not precisely symmetric in morphology), but almost uniform within each, making it 00–. The eigenvalues of the two eigenmodes that emerge from the lowest unihemispheric eigenmode are found to be $\lambda_u^{(0)} = 0.88$ and $\lambda_d^{(0)} = 0.58$, when the deCM normalization is chosen to accord with that used by Robinson et al. (2014). Using Eqs. (60) and (61), these values yield $J \approx 0.73$ and $I \approx 0.15$, which imply that the interhemispheric input to a given region is about a fifth as large as the intrahemispheric one, and comparable to typical extracortical inputs (whose characteristic size is $1 - J - I \approx 0.12$ on this scale.) These relative sizes accord with the fact that intrahemispheric white matter fibers outnumber transcallosal ones severalfold. The eigenvalues of eigenmodes with nonzero λ are not expected to correspond precisely to those shown in Fig. 12 for reasons noted above.

We identify the third to fifth eigenmodes as the 1–1+, 10+, and 11+ eigenmodes, respectively, because each is symmetric between hemispheres and has a single nodal line within each hemisphere; here, we assign μ values in order of energy, as in earlier sections. Although it does not apply exactly for these eigenmodes in the present case, Fig. 12 is consistent in implying that we would expect them to be among the next few to be encountered at $I = 0.15$.

We identify the sixth, eighth, and ninth eigenmodes as the 2–2+, 2–1+, and 20+ eigenmodes on the basis of their symmetry between hemispheres and the two nodal lines seen in each hemisphere, with μ values again assigned in order of energy. These eigenmodes appear significantly out of the order shown in Fig. 12, but there is no contradiction

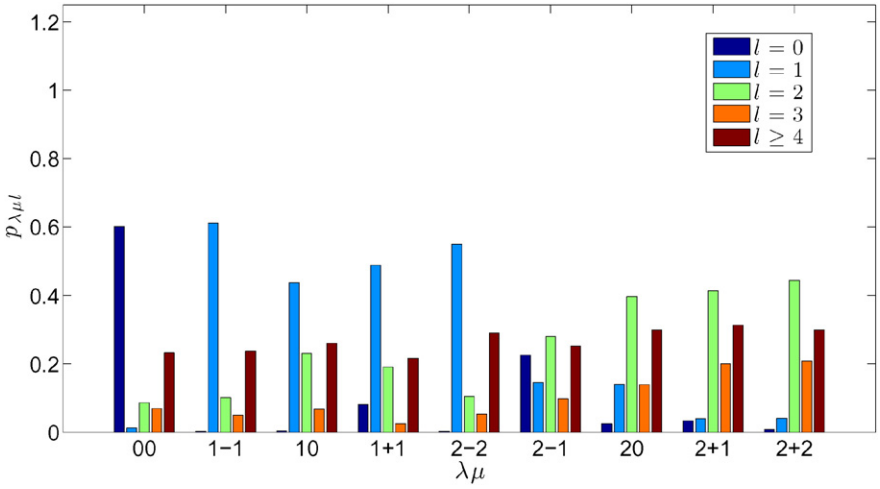


Fig. 11. Spherical harmonic decompositions of the first 9 right-hemisphere experimental deCM eigenmodes from Fig. 9 as measured by the coefficients $p_{\lambda\mu l}$ in Eq. (44), with μ values arbitrarily assigned in order of energy. For each eigenmode $\lambda\mu$, the histogram shows the coefficients for $l = 0, \ldots, 3, 4^+$. The total of each histogram is unity because the spherical harmonics are orthonormal.

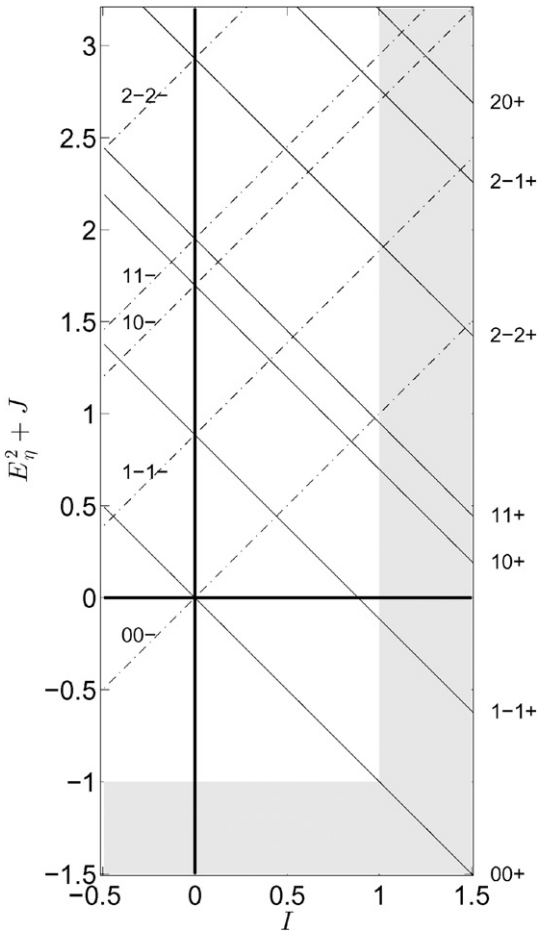


Fig. 12. Mode quantity E_η^2+J vs. the interhemispheric connection strength parameter I. Symmetric and antisymmetric eigenmodes are shown with solid and broken lines, respectively, labeled with their eigenmode number $\lambda\mu\varepsilon$. Eigenvalues at $I=0$ are set to the averages of those for left and right unihemispheric eigenmodes from Figs. 8 and 9, respectively. Unstable regions for $J=0$ are shaded; the stable zone for other J lies above J–1 on the vertical axis and to the left of $I=1-J$.

because Fig. 12 does not take into account the oscillatory frequencies of eigenmodes with $\lambda>0$, as explained earlier.

The seventh eigenmode is identified as the 1–1– eigenmode because it is antisymmetric between hemispheres, has one nodal line in each, and is given the lowest available μ value on the basis of energy.

Notably, Fig. 13 implies an even greater preponderance of symmetric modes than expected from Fig. 12; hence, we infer that inclusion of oscillatory properties increases the energies of antisymmetric modes relative to symmetric ones.

Summary and discussion

We have used neural field theory to calculate eigenmodes of brain activity on the curved cortex, with thalamic interactions, and have compared these with eigenmodes of experimental connection matrices obtained via matrix analysis. The key results follow, along with discussion of their implications and relationship to the literature and the avenues they open up for future work.

(i) We have derived an eigenmode equation for NFT in differential form defined on one hemisphere. This takes the form of the Helmholtz equation, which we have solved numerically on the curved cortical surface to obtain spatial eigenmodes. Each spatial eigenvalue then appears in the corresponding dispersion relation for its eigenmode, a relation that includes the effects of local and subcortical interactions and yields the eigenmode frequency and stability characteristics. The lowest eigenmodes are nonoscillatory, with others having nonzero frequency; the lowest eigenmodes are also the least stable and hence the most easily excited Robinson (2003); Robinson et al. (1997, 2001a). Once the spatial eigenvalue is known, the results of these previous works relating to dispersion and stability can be applied directly.

These results generalize earlier analyses to the case of an arbitrary geometry for a brain hemisphere and extend our preliminary work (Robinson et al., 2015). In the case of a planar geometry, the results reproduce those of Fourier analyses Robinson et al. (1997). In the case of a spherical geometry, the eigenmodes reduce to spherical harmonics. Notably, spherical harmonics have long been used to represent EEG signals and the underlying brain activity (Nunez, 1974; Nunez and Srinivasan, 2006; Wingeier et al., 2001).

(ii) Our numerical solutions of the NFT eigenmode equation confirm the orthonormality of the eigenmodes, and their reduction to spherical harmonics in the appropriate limit. They also show that the eigenmodes preserve the main structural properties and principal-eigenvalue ordering of the spherical harmonics even on the folded cortex. Notably, cortical folding changes the eigenvalues of eigenmodes, but eigenvalues never cross, so families with different angular momentum mode numbers λ remain segregated.

When projected onto a sphere, the spherical harmonic decompositions of NFT eigenmodes with mode numbers $\lambda\mu$ are dominated by spherical harmonics with $l=\lambda$, even for a fully folded cortex. This implies that the spherical harmonics can provide a remarkably good approximation to the structure of the actual eigenmodes, with the advantage that their analytic properties are well known and available for use in further analyses. Symmetry breaking by cortical folding implies that the eigenmodes remain superpositions of multiple spherical harmonics in an arbitrarily chosen coordinate system, even as the degree of folding approaches zero.

(iii) We have used an experimental anatomical CM from the literature to approximate the direct effective CM. The resulting eigenvectors represent the predicted eigenmodes when each element is mapped to the spatial location of the corresponding node, and we also obtain the relevant eigenvalues. The spatial structures of the lowest eigenmodes closely match those of the lowest NFT eigenmodes and all the properties discussed under (ii) apply, except that the dominance of $l=\lambda$ in the spherical harmonic decomposition is not as great, likely due to the effects such as those of data gaps and discretization. This correspondence is despite the CM eigenmodes having been obtained in a completely different manner (eigenvectors calculated by matrix analysis) than in the NFT case (solution of the Helmholtz equation).

The close match between the NFT and deCM eigenmodes raises important questions of why the correspondence is so good and what its implications are for brain structure and activity. Notably, the NFT results are calculated using homogeneous dynamic properties and connectivities (via white matter projections from cortical pyramidal neurons), apart from allowing for cortical folding. In contrast, deCMs are widely thought to contain much structure that is particular to individual brain areas, and to have structure that is predominantly hierarchical, modular, or both, for example (Meunier et al., 2010). The present results imply that, to the extent that it exists, inhomogeneous structure does not have a dominant effect on large-scale brain dynamics. These results are in accord with recent analyses (Henderson and Robinson, 2013, 2014) that emphasized that the bulk of the connections between brain areas are homogeneous in structure — i.e., the distribution of axonal ranges does not depend on location in the brain — with only a few per cent being specific to particular areas. Thus, if area-specific connectivity is to be identified, the large-scale dominance of homogeneous connectivity must first be analyzed and accounted for to unmask it. A further implication is that the NFT spatial propagator is a good approximation to the spatially structured couplings present in the brain and represented in the deCM, a point first emphasized by Robinson et al. (1997).

(iv) We have incorporated the effects of transcallosal couplings between homologous points in opposite hemispheres into NFT in the

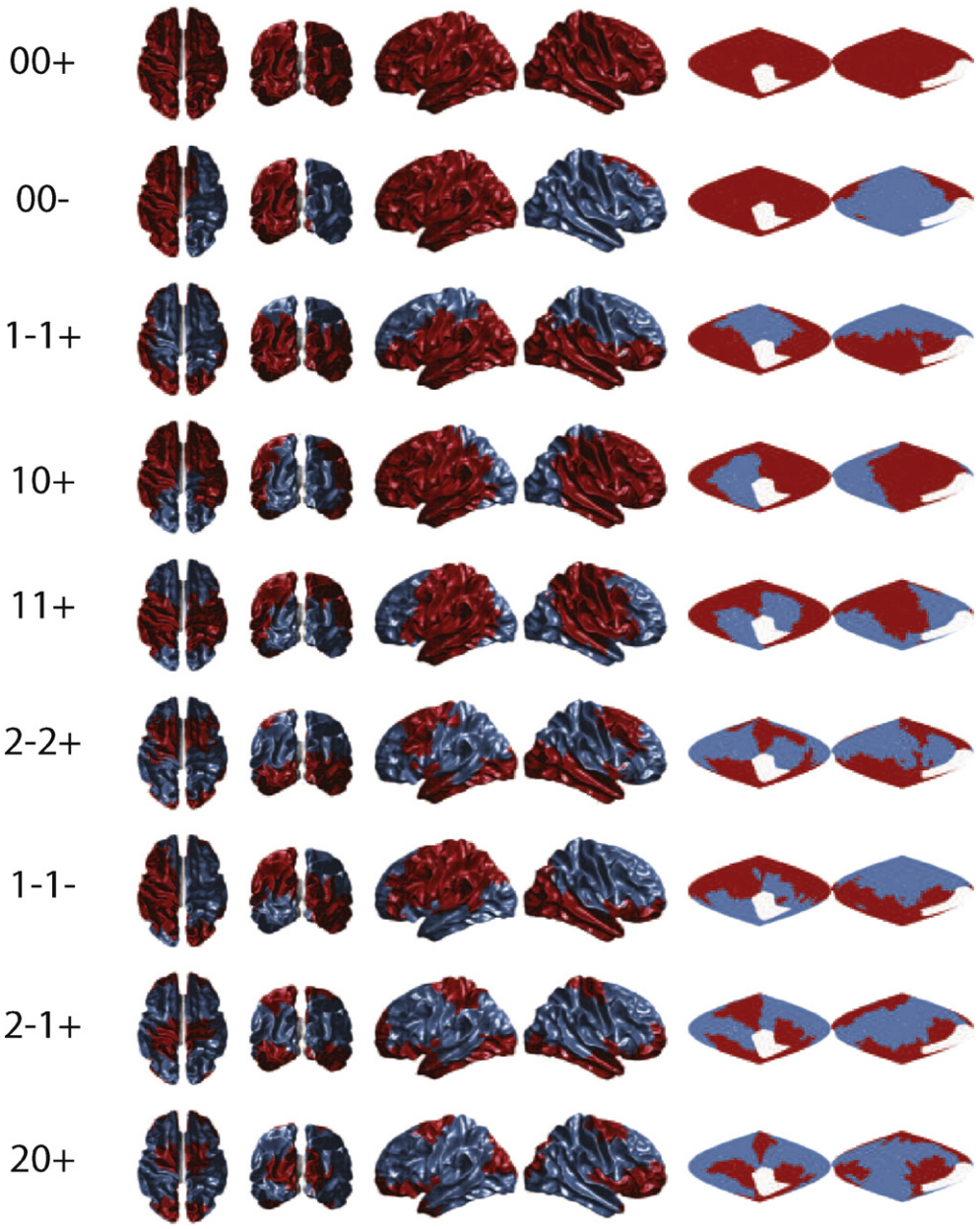


Fig. 13. First 9 eigenmodes of the bihemispheric experimental deCM for the data in the Connectivity matrix and cortical surface data section, starting with the lowest eigenmode at top, ordered by eigenvalue and labeled with mode numbers λμε, as shown. Red indicates positive regions and blue negative ones. In each row the first and second columns show two views of the whole brain, the third and fourth columns show side views of the left and right hemispheres, and the fifth and sixth columns show the corresponding left- and right-hemisphere flattened projections defined in the Flattening of cortical surface onto a plane for visualization section.

approximation of one-to-one coupling. This yields new, bihemispheric eigenmodes that preserve the spatial structure of unihemispheric eigenmodes, but include an additional overall symmetry or antisymmetry between hemispheres. Interhemispheric coupling causes separate left and right unihemispheric eigenmodes to be replaced by symmetric and antisymmetric bihemispheric ones. When the interhemispheric coupling is excitatory, as expected, the symmetric eigenmode of each pair has a lower energy than the antisymmetric one, and is thus less stable.

Some implications of these results are that the eigenmodes can cross in eigenvalue if the interhemispheric coupling is sufficiently strong, and that the sign of this coupling can be determined by examining the symmetry of the lowest eigenmode. Notably, for positive coupling, the preponderance of symmetric eigenmodes among those of lowest energy can immediately account for much of the observed dominance of symmetric brain activity, which is likely to be reinforced by the natural preponderance of symmetric inputs.

(v) Determination of eigenvectors from an experimental deCM from the literature confirms the above results from NFT. The eigenmodes are either symmetric or antisymmetric between hemispheres, and preserve the unihemispheric eigenmode structure, closely resembling spherical harmonics within each hemisphere. The lowest eigenmode is uniform within each hemisphere and symmetric between them, thereby implying excitatory interhemispheric couplings.

(vi) The observed splitting between the $00+$ and $00-$ eigenmodes implies that the intrahemispheric coupling coefficient is $J \approx 0.73$ and the interhemispheric coupling is $I \approx 0.15$. On the future experimental front, higher-frequency measurements via EEG or MEG would enable more detailed comparison of the higher eigenvalues with those predicted by NFT. It would also be telling to compare the theory with data from

acallosal subjects, or subjects with reduced callosal function due to lesions or partial agenesis, to determine relative interhemispheric coupling strengths and intrahemispheric eigenmode structure — one would expect less dominance of symmetric eigenmodes, for example.

(vii) Turning to wider discussion of the results, we first note that the present work demonstrates a remarkable level of agreement between NFT predictions and experimental estimates of eigenmode structure and properties. Perfect agreement is not expected, owing to the approximations made in the theoretical work and the imperfections of the experimental data, but the results are a major advance on the pre-existing situation, in which comparison was not even possible.

A number of the approximations can be investigated further and/or improved upon. Notably, we have not investigated integral forms of NFT that do not have exact differential counterparts. Similarly, sources of eigenmode coupling and structure other than folding should also be incorporated, including spatial variation of the physiological parameters and asymmetries between hemispheres. Exploration of higher frequencies should be carried out in more detail, generalizing modal results such as those of Robinson et al. (2001a); Robinson (2003) to the curved cortex, and including asymmetries in NFT.

On the experimental side, more work needs to be done on increasing the density and uniformity of sampling, avoiding missed interhemispheric connections, and improving the handling of residual data gaps especially where the corpus callosum passes between hemispheres. Calculation of connectivities based on correlation functions is also needed before modes on experimental CMs can be compared with theoretical ones at frequencies above about 1 Hz, because time delays and asymmetries become critical to determining activity patterns in this range (Robinson et al., 2002).

All these improvements notwithstanding, our key theoretical predictions, and their agreement with experiment, are expected to remain robust — most notably modal structure and symmetry properties between hemispheres — since these rest on fundamental properties that do not depend strongly on model details or experimental refinements.

(viii) The present work provides new perspectives on a wide range of phenomena, likely resolving many of the issues discussed in the Introduction. Most notably, it recognizes that brain activity is concentrated in eigenmodes, rather than discrete "networks", and these eigenmodes provide the discrete patterns that are seen in a wide variety of experiments. This is directly analogous to how a continuous violin string, for example, produces a set of discrete notes, each with a distinct spatial pattern of vibration, but common across all pieces of music in which it can be played. A given point on the string takes part in all notes, and no spatial pattern is due to a separate "network" of points on the string — all points are needed to generate each note.

This picture immediately accounts for the similarities between patterns of brain activity seen in spontaneous and task conditions; the same eigenmodes are excited by different drivers and in different proportions. Changes in neural gains affect the modal dispersion, but not the spatial eigenmode structure, thereby also explaining the observed similarity of patterns seen in conscious and unconscious states. Since eigenmodes have specific frequency-structure relationships, via their dispersion relations, they can also potentially account for frequency dependent connectivity properties (Kitzbichler et al., 2009), although we have not explored high frequencies explicitly in this paper.

The similarity of eigenmodes on spherical and folded surfaces explains why only relatively few eigenmodes are robustly seen in experimental data: each time λ increases by one an additional nodal line is added in the spherical limit, dividing the cortical surface into a rapidly increasing number of zones. The broad features of this structure translate directly into the folded case, although some exact nodal line crossings are eliminated. Hence, the characteristic scales involved rapidly decrease and the signal-to-noise ratio available to detect a given zone also diminishes. Additionally, eigenmodes may be rotated by amounts that vary between subjects, as is freely allowed on the sphere (albeit less so on the brain, due to symmetry breaking). This effect makes it more difficult to match observed patterns between individuals, especially for higher eigenmodes with more complex structure, and will reduce signal-to-noise ratios when averaged over subjects. However, spherical-harmonic decomposition can enable dominant structure to be projected out in a way that is less subject to these problems. We note that the roughly 5–10 patterns typically reported in the literature correspond to $\lambda \leq 2$, where there are at most 4 zones in (the spherical limit of) any modal pattern; $\lambda = 3$ eigenmodes have 6–8 zones each.

(ix) A modal perspective immediately accounts for why activity in some pairs of brain areas is anticorrelated — eigenmodes naturally have positive and negative zones and, likewise, high-amplitude and low-amplitude regions. Thus, excitation of a particular eigenmode will tend to anticorrelate amplitudes in different areas, and likewise to anticorrelate powers relative to the mean.

(x) Modal methods enable new approaches to many problems. For example, instead of seeking implausibly discrete "networks" to explain different patterns of activity, these patterns are seen to be different excitations of the same neural tissue. This type of perspective has been massively fruitful across numerous fields of physics in the last two centuries, in areas ranging from heat, acoustics, and electromagnetics to quantum mechanics and elementary particle physics. The immensely powerful tools developed in those disciplines can now be brought to bear on problems in brain dynamics and connectivity.

The question of "resting state" and "default mode" "networks" can also be addressed from this viewpoint. Usually, these are determined by methods such as ICA, which seek to find a minimal number of causal patterns to account for observed data. From a modal perspective, these analyses are seen to be seeking dominant combinations of drives and corresponding modal responses, rather than eigenmodes alone. Some of these methods even use empirical activity eigenmodes as an intermediate analytic step, but they are forms of blind signal analysis that do not exploit the physiological underpinnings of the system (and thus are not able to be directly related to physiology), nor the full power of using the eigenmodes that are the intrinsic dynamical entities of the system. In the abovementioned branches of physics, the dynamical reality of eigenmodes adds analytic power well beyond that provided by blind signal analyses, although the latter remain useful if physically based system models are not available.

(xi) Modal perspectives point the way to improved data analyses. For example, one could use observed eigenmodes as regressors to project out the amplitudes of fMRI signals in various dynamically relevant oscillations and spatial patterns, rather than employing regressors of more arbitrary structure. Because NFT establishes relationships between modal excitation by stimuli, it can also potentially aid in removal of global artifacts, such as heartbeat and breathing, which can affect the covariances usually used to define functional CMs.

(xii) Experimental design and analysis can also benefit from modal analysis. For example, if one wishes to excite a particular eigenmode, a stimulus should be chosen whose location on the cortex corresponds to a peak of that eigenmode, not to one of its nodal lines. This point is closely related to optimal choice of regions of interest to include in analyses such as dynamic causal modeling (Robinson, 2013). The effect of seed location on seed-based correlation analyses can also be analyzed in terms of eigenmodes. Interestingly, because modes are pinned to specific orientations on the individual folded cortex by symmetry breaking, seeds (or stimuli in task experiments) that are not located exactly at the peak of a given eigenmode will still tend to yield similar patterns of correlations (or activation), with the same eigenmodes, implying that connectivity should not display extreme sensitivity to the seed or stimulus location.

(xiii) In the linear regime, eigenmodes are completely decoupled from one another, so an eigenmode-eigenmode CM will be exactly diagonal — an extremely sparse representation of connectivity that explicitly exhibits the fact that $n \times n$ CMs can be expressed in terms of just n independent eigenmode amplitudes (Robinson et al., 2014; Robinson and Roy, 2015). Moderate nonlinear coupling will introduce

eigenmode-eigenmode interactions which are expected to be expressible in a relatively compact form starting from this initial sparse representation.

(xiv) An exact parallel can be drawn with eigenmodes in other areas of physics: modal descriptions and coordinate-based ones (e.g., in terms of localized regions of interest) are complementary, not mutually exclusive. Some phenomena are best described in local terms, others in terms of delocalized eigenmodes. Both approaches ultimately lead to the same end results, but one usually gives far more direct insight than the other in a given situation. To extract maximum understanding, one must thus move between the two descriptions as the physical situation demands.

(xv) On a speculative note, one can ask whether dynamical eigenmodes have a cognitive or similar role, just as many investigators have speculated on the role of RSNs discovered by statistical means. They may, of course, be an epiphenomenon, but would still have diagnostic value. Alternatively, given that they embody large-scale activity in the cortex, they may play a role in intra- and interhemispheric communication (i.e., broadcast) of macroscopic brain states, cortical homeostasis, and similar phenomena. Their oscillatory nature may also help to enhance synchrony and processes such as plasticity, whose efficiency is increased by synchronous inputs. Similarly, proposals such as communication through coherence (Fries, 2005) may well be facilitated by the existence of such coherent eigenmodes. Certainly, evolution is likely to have exploited such properties if this enhanced fitness.

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